

INTRODUCTION TO THE CELL

Cells, Genomes, and the Diversity of Life

CHAPTER

1

The surface of our planet is populated by living things—*organisms*—curious, intricately organized chemical factories that take in matter from their surroundings and use these raw materials to generate copies of themselves. These organisms appear extraordinarily diverse. What could be more different than a tiger and a piece of seaweed or a butterfly and a tree? Yet our ancestors, knowing nothing of cells or DNA, saw that all these things had something in common. They called that something “life,” marveled at it, struggled to define it, and despaired of explaining what it was or how it worked in terms that relate to non-living matter.

The remarkable discoveries of the past 100 years or so have not diminished the marvel—quite the contrary. But they have removed the central mystery regarding the nature of life. We can now see that all living things are made of cells: small, membrane-enclosed units filled with a concentrated aqueous solution of chemicals and endowed with the extraordinary ability to create copies of themselves by growing and then dividing in two.

Because cells are the fundamental units of life, it is to *cell biology*—the study of the structure, function, and behavior of cells—that we must look for answers to the questions of what life is and how it works. With a deeper understanding of cells and their evolution, we can begin to tackle the grand historical problems of life on Earth: its mysterious origins, its stunning diversity, and its invasion of every conceivable habitat. Indeed, as emphasized long ago by the pioneering cell biologist E. B. Wilson, “the key to every biological problem must finally be sought in the cell; for every living organism is, or at some time has been, a cell.”

Despite their apparent diversity, living things are fundamentally similar inside. The whole of biology is thus a counterpoint between two themes: astonishing variety in individual particulars and astonishing constancy in fundamental mechanisms. In this chapter, we begin by outlining the universal features common to all life on our planet, along with some of the fundamental properties of their cells. We then discuss how an analysis of DNA *genomes* allows scientists to position the wide variety of organisms in an evolutionary “tree of life.” This

IN THIS CHAPTER

The Universal Features of Life on Earth

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approach, which quantifies how closely organisms are related to one another, allows us to identify the three major branches of life on Earth, *eukaryotes*, *bacteria*, and *archaea*—each with unique qualities. We shall see that the familiar world of plants and animals—the focus of scientists for many centuries—makes up only a small slice of the complete diversity of life, the vast majority of which is invisible to the unaided human eye.

After exploring some of the ways that genomes change over evolutionary times, we highlight the handful of *model organisms* that biologists have chosen to focus on to dissect the molecular mechanisms underlying life. A few specific viruses, including SARS-CoV-2, pose grave threats to humans, so they too have become objects of intensive study. For this reason, this section also includes an introduction to viruses, the ubiquitous parasites that have evolved to feed on cells. Viruses are now recognized to be the most abundant biological entities on the planet.

THE UNIVERSAL FEATURES OF LIFE ON EARTH

There are more than 2 million described species living on Earth today, but many, many more are yet to be discovered. Each species is different, and each reproduces itself faithfully, yielding progeny that are unique to that species. Thus, the parent organism hands down information specifying, in extraordinary detail, the characteristics that the offspring will have. This phenomenon of *heredity* is central to the definition of life: it distinguishes life from other processes, such as the growth of a crystal, or the burning of a candle, or the formation of waves on water, in which structures are generated without the same type of link between the peculiarities of parents and offspring. A living organism must consume free energy to exist, as does a candle flame. But life employs this *free energy* to drive a very complex system of chemical reactions that create and maintain the intricate organization of its cells, all as specified by the hereditary information in those cells.

Most living organisms are single cells. Others, such as us, are like vast multicellular cities in which groups of cells perform specialized functions that are linked by intricate systems of intercellular communication. But even for the aggregate of more than 10^{13} cells that makes up a human body, the whole organism has been generated by cell divisions from a single cell. The single cell therefore contains all of the hereditary information that defines a species (Figure 1-1). The cell must also contain all of the machinery needed to gather raw materials from the environment and to construct from them a new cell in its own image, complete with a new copy of the hereditary information of its parent. Every cell on Earth is truly amazing.

All Cells Store Their Hereditary Information in the Form of Double-Strand DNA Molecules

Computers have made us familiar with the concept of information as a measurable quantity— 10^6 bytes to record a few hundred pages of text or an image from a digital camera, 10^9 bytes for a 60-minute video streamed from the Internet, and so on. Computers have also made us well aware that the same information can be recorded in many different physical forms: the discs and tapes that we used 25 years ago for our electronic archives have become unreadable on present-day machines. Living cells, like computers, store information, and it is estimated that they have been evolving and diversifying for more than 3.5 billion years. One might not expect that they would all store their information in the same form or that the hereditary information carried by one type of cell should be readable by the information-handling machinery of another. And yet it is so. This fact provides compelling evidence that all living things on Earth have inherited the form of their genetic instructions, as well as how to use them, from a *universal common ancestral cell*. This ancestor is thought to have existed roughly 3.5–3.8 billion years ago.

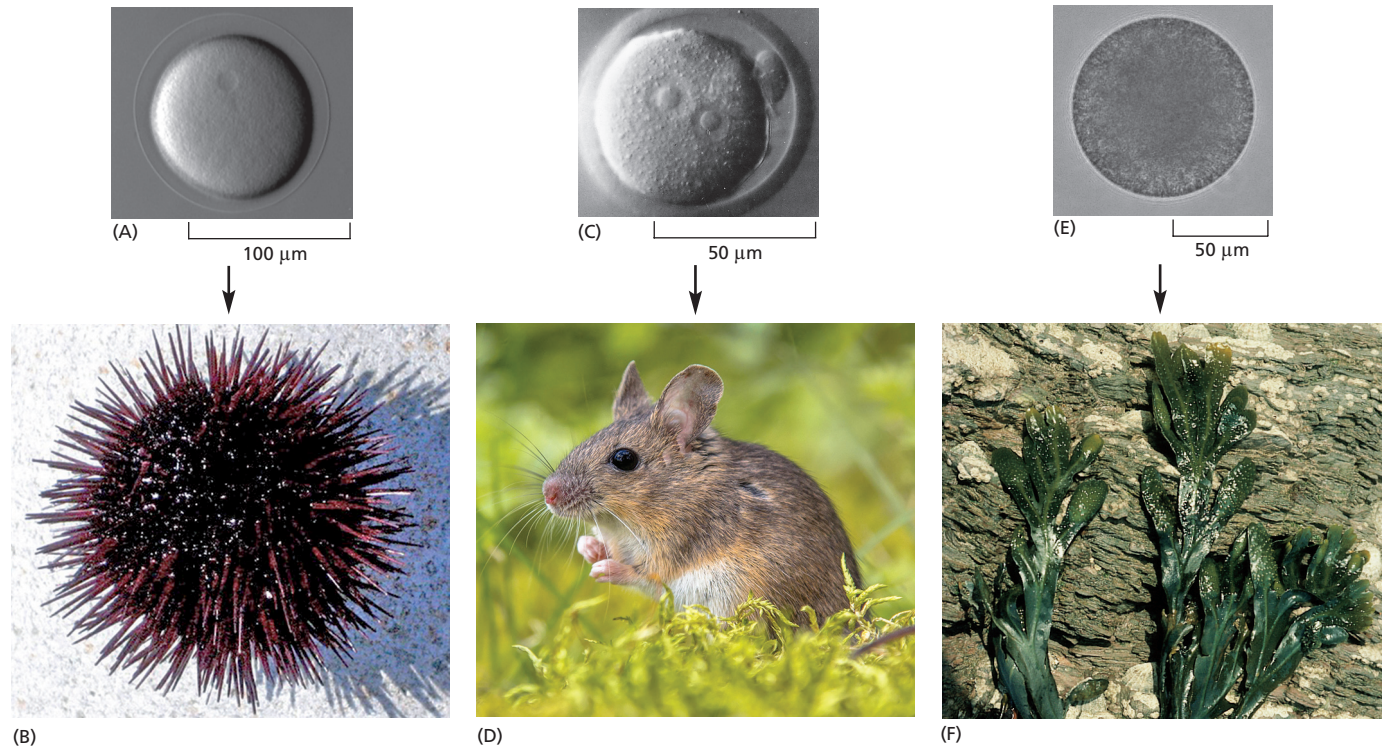


Figure 1–1 The hereditary information in the fertilized egg cell determines the nature of the whole multicellular organism that will develop from it. As indicated, although their starting cells look superficially similar, the egg of a sea urchin gives rise to a sea urchin (A and B), the egg of a mouse gives rise to a mouse (C and D), and the egg of the seaweed *Fucus* gives rise to a *Fucus* seaweed (E and F). (A, courtesy of David McClay; B, courtesy of Tim Hunt; C, courtesy of Patricia Calarco, from G. Martin, *Science* 209:768–776, 1980. With permission from AAAS; D, Rudmer Zwerver/Alamy Stock Photo; E and F, courtesy of Colin Brownlee.)

All cells on Earth today store their hereditary information in the form of double-strand molecules of DNA—long, unbranched, paired *polymer* chains, which are always composed of the same four types of *monomers*. These monomers, chemical compounds known as *nucleotides*, have nicknames drawn from a four-letter alphabet—A, T, C, G—and they are strung together in a long linear sequence that encodes the hereditary information, just as the sequence of 1's and 0's encodes the information in a computer file. We can take a piece of DNA from a human cell and insert it into a bacterium or a piece of bacterial DNA and insert it into a human cell, and, with only a few minor modifications, the information will be successfully read, interpreted, and copied. As we describe in Chapter 8, scientists can now rapidly read out the sequence of nucleotides in any DNA molecule and thereby determine the complete DNA sequence of any cell's **genome**—the totality of its hereditary information embodied in the linear sequence of nucleotides in its DNA. As a result, we now know the complete genome sequences for tens of thousands of species, ranging from the smallest bacterium to the largest plants and animals on Earth.

All Cells Replicate Their Hereditary Information by Templated Polymerization

The mechanisms that make life possible depend on the structure of the double-strand DNA molecule. We discuss this remarkable molecule in detail in Chapters 4 and 5; here we provide only an overview of its structure and means of reproduction. Each monomer in a single DNA strand—that is, each **nucleotide**—consists of two parts: a sugar (deoxyribose) with a phosphate group attached to it, and a *base*, which may be either adenine (A), guanine (G),

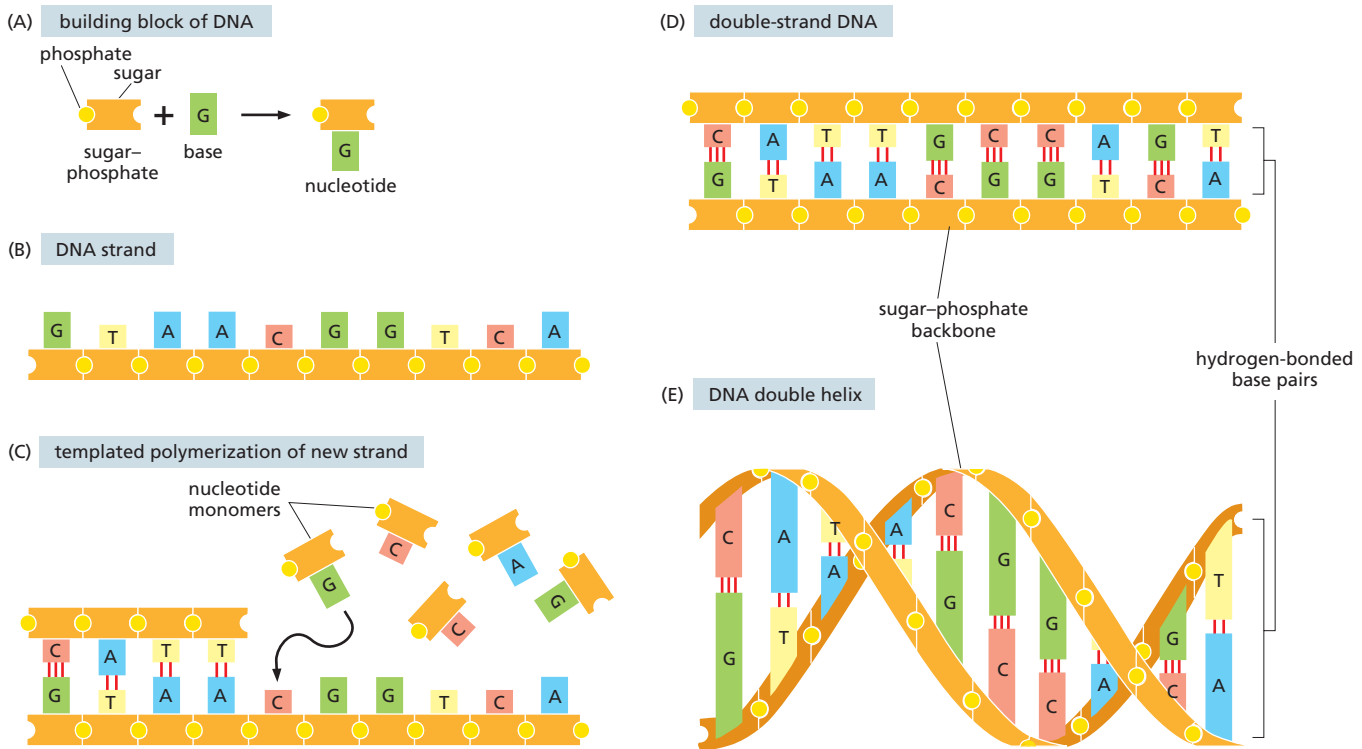


Figure 1-2 DNA and its building blocks. (A) DNA is made from simple subunits, called nucleotides. Each nucleotide consists of a specific arrangement of about 35 covalently linked atoms, forming a sugar-phosphate molecule with a nitrogen-containing side group, or base, attached to it. The bases are of four types (adenine, guanine, cytosine, and thymine), corresponding to four distinct nucleotides, labeled A, G, C, and T. (B) A single strand of DNA consists of nucleotides joined together by sugar-phosphate linkages. Note that the individual sugar-phosphate units are asymmetric, giving the backbone of the strand a definite directionality, or polarity. This directionality guides the molecular processes by which the information in DNA is both interpreted and copied (*replicated*) in cells: the information is always “read” in a consistent order, just as written English text is read from left to right. (C) Through templated polymerization, the sequence of nucleotides in an existing DNA strand controls the sequence in which nucleotides are joined together in a new DNA strand; T in one strand pairs with A in the other, and G in one strand with C in the other. The new strand therefore has a nucleotide sequence *complementary* to that of the old strand and a backbone with opposite directionality: thus, GTAA... in the original strand, is ...TTAC in the new strand. (D) A normal DNA molecule consists of two such complementary strands. The nucleotides within each strand are linked by strong (covalent) chemical bonds; the complementary nucleotides on opposite strands are held together more weakly, by hydrogen bonds. (E) The two strands twist around each other to form a double helix—a robust structure that can accommodate any sequence of nucleotides without altering its basic double-helical structure (see Movie 4.1).

cytosine (C), or thymine (T) (Figure 1-2). Each sugar is linked to the next via the phosphate group, creating a polymer chain composed of a repetitive sugar-phosphate backbone with a series of bases protruding from it. The DNA polymer is extended by adding monomers at one end. For a single isolated strand, these monomers can, in principle, be added in any order, because each one links to the next in the same way, through the part of the molecule that is the same for all of them. In the living cell, however, DNA is not synthesized as a free strand in isolation, but on a template formed by a preexisting DNA strand. The bases that protrude from this template can bind to bases of the strand being synthesized, according to a strict rule defined by the complementary structures of the bases: A binds to T, and C binds to G. This *base-pairing* holds fresh monomers in place and thereby controls the selection of which one of the four monomers will next be added to a growing strand. In this way, a double-strand structure is created, consisting of two exactly complementary sequences of A's, C's, T's, and G's. These two strands twist around each other, forming a DNA double helix (see Figure 1-2E).

Compared with the covalent sugar-phosphate bonds, the hydrogen bonds between the base pairs are weak, which allows the two DNA strands to be

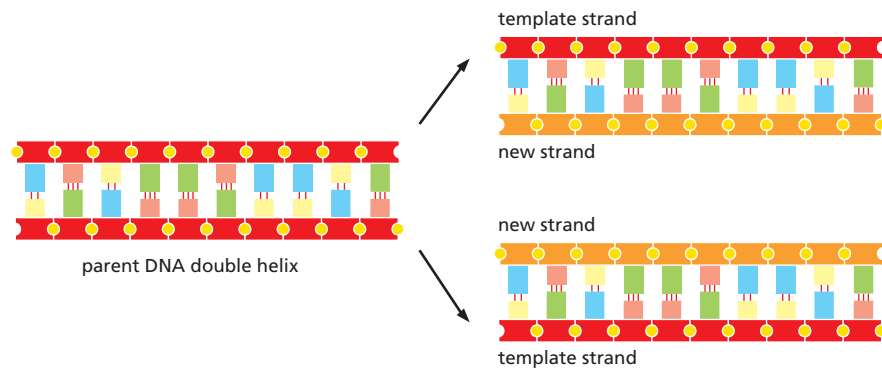


Figure 1-3 The copying of genetic information by DNA replication. In this process, the two strands of a DNA double helix are pulled apart, and each serves as a template for the synthesis of a new complementary strand. The end result is two daughter DNA double helices that are identical in sequence to the parent double helix.

pulled apart without breakage of their backbones. Each strand then can serve as a template, in the way just described, for the synthesis of a fresh DNA strand complementary to itself—a fresh copy, that is, of the hereditary information (**Figure 1-3**). In different types of cells, this process of **DNA replication** occurs at different rates, with different controls to start it or stop it, and with different auxiliary molecules to help the process along (discussed in Chapters 5 and 17). But the basics are universal: DNA is the information store for heredity, and *templated polymerization* is the way in which this information is copied throughout the living world.

All Cells Transcribe Portions of Their DNA into RNA Molecules

To carry out its information-bearing function, DNA must do more than copy itself. It must also *express* its information, by letting the information guide the synthesis of other molecules in the cell. This expression occurs by a mechanism that is the same in all living organisms, leading first and foremost to the production of two other crucial classes of biological polymers: RNA molecules and protein molecules. The process begins with a templated polymerization called **transcription**, in which segments of the DNA sequence are used as templates for the synthesis of shorter molecules of the closely related polymer **ribonucleic acid, or RNA**. Subsequently, in a process called **translation**, many of these RNA molecules direct the synthesis of polymers of a radically different chemical class—the *proteins* (**Figure 1-4**). The detailed chemical reactions involved are presented in Chapter 6; here they will only be briefly outlined.

The backbone of an RNA molecule is formed by a slightly different sugar from that in DNA—ribose instead of deoxyribose; in addition, one of the four bases is slightly different—uracil (U) replaces thymine (T). Most important, however, the other three bases—A, C, and G—are identical to those in DNA, and all four bases will pair with their complementary counterparts in DNA—the A, U, C, and G of RNA with the T, A, G, and C of DNA, respectively. During transcription, this pairing allows the RNA monomers to be lined up and selected for polymerization on a template strand of DNA, just as DNA monomers are selected during replication. The outcome is a single-strand polymer molecule whose sequence of nucleotides faithfully represents a portion of the cell's genetic information, even though it is written in a slightly different alphabet—consisting of the four RNA monomers instead of the four DNA monomers.

The same segment of DNA can be used repeatedly to guide the synthesis of many identical RNA molecules. Thus, whereas the cell's archive of genetic information in the form of DNA is fixed and sacrosanct, *RNA transcripts* are mass-produced and disposable. Most of these transcripts function as intermediates in the transfer of genetic information by serving as *messenger RNA (mRNA)* molecules that guide the synthesis of proteins according to the genetic instructions stored in the DNA. But as we discuss in Chapter 6, some RNA transcripts do not serve as information carriers; instead, they function directly in the cell to carry out a variety of other functions.

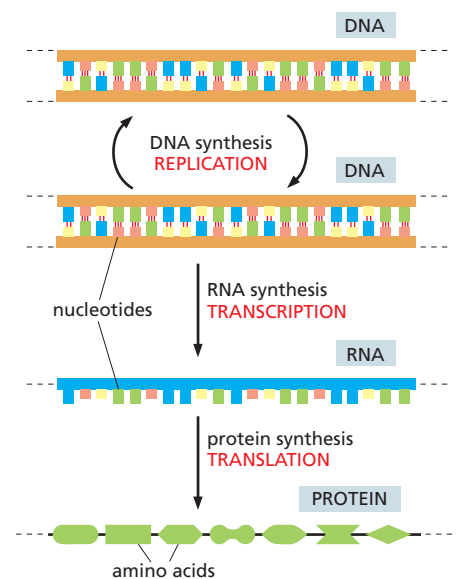


Figure 1-4 From DNA to protein. In addition to DNA replication (shown at the top of the figure), genetic information is read out and put to use through a two-step process: First, in *transcription*, segments of the DNA sequence are used to guide the synthesis of molecules of RNA. Then, in *translation*, RNA molecules are used to guide the synthesis of proteins, which are polymers made of amino acid subunits (discussed shortly).

All Cells Use Proteins as Catalysts

Like DNA and RNA molecules, **protein** molecules are long unbranched polymer chains, formed by stringing together monomeric building blocks (subunits) drawn from a standard repertoire that is the same for all living cells. Like DNA and RNA, proteins carry information in the form of a linear sequence of subunits in the same way as a human message written in an alphabetic script. There are many different protein molecules in each cell, and—if we ignore water molecules—they form the major portion of the cell's mass.

The subunits of proteins are the **amino acids**, which are quite different from the nucleotides of DNA and RNA, and there are 20 types instead of 4. Each amino acid is built around a core structure that allows it to be covalently linked in a standard way to any other amino acid in the set; attached to this core is a side group of atoms that gives each amino acid a distinctive chemical character. Each protein molecule is a *polypeptide* chain that is created by joining its amino acids in a particular sequence; this sequence determines how the polypeptide folds up, giving the protein its unique three-dimensional structure. Through several billion years of evolution, these sequences have been selected to give each protein a useful function.

By folding into a precise structure that binds with high specificity to other molecules, each protein performs a specific function according to its genetically specified sequence of amino acids. Proteins form and maintain diverse cell and extracellular structures, generate movements, sense signals, and so on. Many have reactive sites on their surface, allowing them to act as *enzymes* that catalyze reactions that make or break specific covalent bonds. Proteins, above all, are the main molecules that put the cell's genetic information into action. Thus, polynucleotides (DNA and mRNAs) specify the amino acid sequences of proteins. Proteins, in turn, serve as *catalysts* to cause many different chemical reactions to occur, including those that synthesize new DNA and RNA molecules.

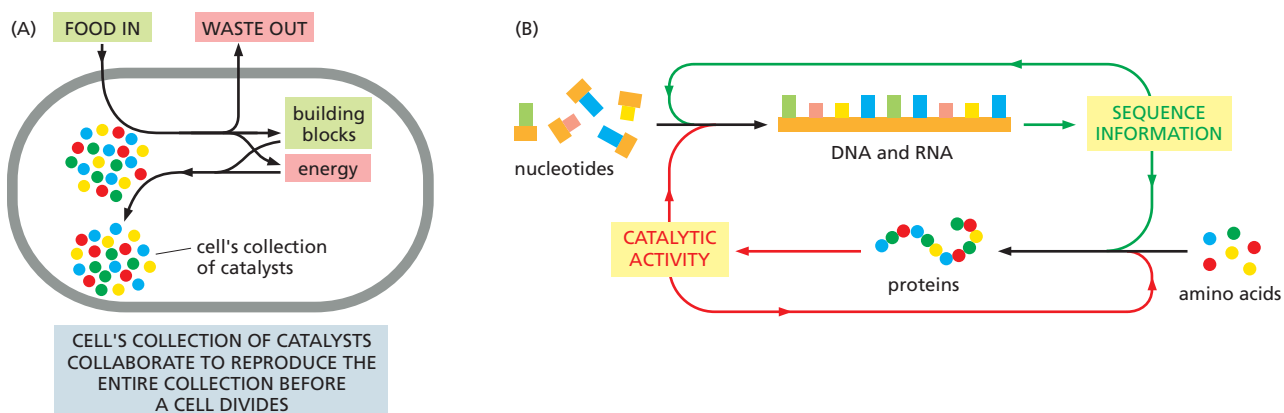
In everyday speech, a catalyst refers to “any agent that provokes or speeds significant change or action.” But in chemistry, the term **catalyst** is defined more narrowly, being applied to any molecule that speeds up a specific chemical reaction without itself being changed. From the most fundamental point of view, a living cell is a self-replicating collection of catalysts that takes in food, processes this food to provide both the building blocks and energy needed to make more catalysts, and discards the materials left over as waste (**Figure 1-5A**). Together, these feedback loops that connect proteins and polynucleotides form the basis for this autocatalytic, self-reproducing behavior of all living organisms (**Figure 1-5B**).

All Cells Translate RNA into Protein in the Same Way

How the information in DNA specifies the production of proteins was a complete mystery in the 1950s when the double-strand structure of DNA was first revealed as the basis of heredity. But in subsequent years, scientists discovered the elegant

Figure 1-5 Life as an autocatalytic process.

(A) The living cell is a self-replicating collection of catalysts. (B) Life can be viewed as an autocatalytic process. DNA and RNA molecules provide the nucleotide sequence information (*green arrows*) that is used both to produce proteins and to copy themselves. Proteins, in turn, provide the catalytic activity (*red arrows*) needed to synthesize DNA, RNA, and proteins themselves. Together, these feedback loops create the self-replicating system that endows cells with the ability to reproduce. Although the great majority of the catalysts in the cell are proteins (known as enzymes), a few RNA molecules (known as ribozymes) also have this property, as we will see in Chapter 6.



mechanisms involved. The translation of genetic information from the 4-letter alphabet of polynucleotides into the 20-letter alphabet of proteins is a complex process. The rules of this translation seem in some respects neat and rational but in other respects strangely arbitrary, given that they are (with minor exceptions) identical in all living things. These arbitrary features, it is thought, reflect frozen accidents in the early history of life. They stem from the chance properties of the earliest organisms that were passed on by heredity and have become so deeply embedded in the constitution of all living cells that they cannot be changed without disastrous consequences.

It turns out that the information in the sequence of a messenger RNA (mRNA) molecule is read out in groups of three nucleotides at a time: each triplet of nucleotides, or *codon*, specifies (codes for, or encodes) a single amino acid in a corresponding protein. Because the number of distinct triplets that can be formed from four nucleotides is 4^3 , there are 64 possible codons, all of which occur in nature. However, there are only 20 naturally occurring amino acids, which means there are necessarily many cases in which several codons correspond to the same amino acid. This *genetic code* is read out by a special class of small RNA molecules, called *transfer RNAs* (tRNAs). Each type of tRNA becomes attached at one end to a specific amino acid and displays at its other end a specific sequence of three nucleotides—an *anticodon*—that enables it to recognize, through base-pairing, a particular codon or subset of codons in mRNA. The intricate chemistry that enables these tRNAs to translate a specific sequence of A, C, G, and U nucleotides in an mRNA molecule into a specific sequence of amino acids in a protein molecule occurs on a *ribosome*, a large multimolecular machine composed of both protein and *ribosomal RNA*. All of these processes will be described in detail in Chapter 6.

Each Protein Is Encoded by a Specific Gene

DNA molecules as a rule are very large, containing the specifications for thousands of proteins and RNA molecules. Special sequences in the DNA serve as punctuation, defining where the information for each RNA and protein begins and ends. And individual segments of the long DNA sequence are transcribed into separate mRNA molecules, coding for different proteins. Each such DNA segment represents one gene. As previously mentioned, some DNA segments—a smaller number—are transcribed into RNA molecules that are not translated into protein but have other functions in the cell; such DNA segments also count as genes. A **gene** therefore is defined as the segment of DNA sequence corresponding either to a single protein (but sometimes to a set of closely related, alternative protein variants) or to a single catalytic, regulatory, or structural RNA molecule.

In all cells, the *expression* of individual genes is regulated: instead of manufacturing its full repertoire of possible proteins and RNAs at full tilt all the time, the cell adjusts the rate of transcription and translation of different genes independently, according to need. As we shall see in Chapter 7, stretches of *regulatory DNA* are interspersed among the segments that code for protein, and these noncoding regions bind to special protein molecules that control the rate of transcription of individual genes. The organization of this regulatory DNA varies widely from one class of organisms to another, but the basic strategy is universal. In this way, the genome of the cell dictates not only the nature of the cell's proteins but also when and where they are to be made.

Life Requires a Continual Input of Free Energy

A living cell is a dynamic chemical system, operating far from chemical equilibrium. For a cell to grow or to make a new cell in its own image, it must take in *free energy* from the environment, as well as raw materials, to drive the necessary synthetic reactions. This consumption of free energy is fundamental to life. When this energy is not available, a cell decays toward chemical equilibrium and soon dies.

As one example, free energy is required for the propagation of genetic information. Picture the molecules in a cell as a swarm of objects endowed with thermal energy, moving around violently at random, buffeted by collisions with one another. To copy genetic information—in the form of a DNA sequence, for example—nucleotide molecules from this wild crowd must be captured, arranged in a specific order defined by a preexisting template, and linked together in a fixed relationship. The bonds that hold the nucleotides in their proper places on the template and join them together must be strong enough to resist the disordering effect of thermal motion, which we describe shortly. The joining process is driven forward by a consumption of free energy, which is needed to ensure that the correct bonds are made, and made robustly. As an analogy, the molecules might be compared with spring-loaded traps, ready to snap into a more stable, lower-energy attached state when they meet their proper partners. As they snap together into the bonded arrangement, their available stored energy—their free energy—like the energy of the spring in the trap, is released and dissipated as heat. In a cell, the chemical processes underlying information transfer are more complex, but the same basic principle applies: free energy must be spent for the creation of order.

To replicate its genetic information faithfully, and indeed to make all its complex molecules according to the correct specifications, the cell therefore requires free energy, which has to be imported somehow from the surroundings. As we will discuss in detail in Chapter 2, the free energy required by animal cells is derived from chemical bonds in food molecules that the animals eat, whereas plants get their free energy from sunlight.

All Cells Function as Biochemical Factories

Because all cells make DNA, RNA, and protein, they all have to contain and manipulate a similar collection of small organic (carbon-containing) molecules, including simple sugars, nucleotides, and amino acids, as well as other substances that are universally required. All cells, for example, require the phosphorylated nucleotide *ATP* (*adenosine triphosphate*), not only as a building block for the synthesis of DNA and RNA but also as a carrier of the free energy that is needed to drive a huge number of chemical reactions in the cell.

Although all cells function as biochemical factories of a broadly similar type, many of the details of their small-molecule transactions differ. Plants, for example, require only the simplest of nutrients because they harness the energy of sunlight to make all their own small organic molecules. Other organisms, such as animals and some bacteria, feed on living (or once living) organisms and must obtain many of their organic molecules ready-made. We return to this point later in the chapter.

All Cells Are Enclosed in a Plasma Membrane Across Which Nutrients and Waste Materials Must Pass

Each living cell is enclosed by a membrane—the **plasma membrane**. This membrane acts as a selective barrier that enables the cell to concentrate nutrients gathered from its environment and retain the products it synthesizes for its own use, while excreting its waste products. Without a plasma membrane, the cell could not maintain its integrity as a coordinated chemical system.

The molecules that form cell membranes have the simple physicochemical property of being *amphiphilic*; that is, they consist of one part that is hydrophilic (water-soluble) and another part that is hydrophobic (water-insoluble). Such molecules placed in water aggregate spontaneously, arranging their hydrophobic portions to be as much in contact with one another as possible to hide them from the water, while keeping their hydrophilic portions exposed. Amphiphilic molecules of appropriate shape, such as the *phospholipid* molecules that compose most of the molecules of the plasma membrane, spontaneously aggregate in water to create a *bilayer* that forms small closed vesicles (Figure 1–6).

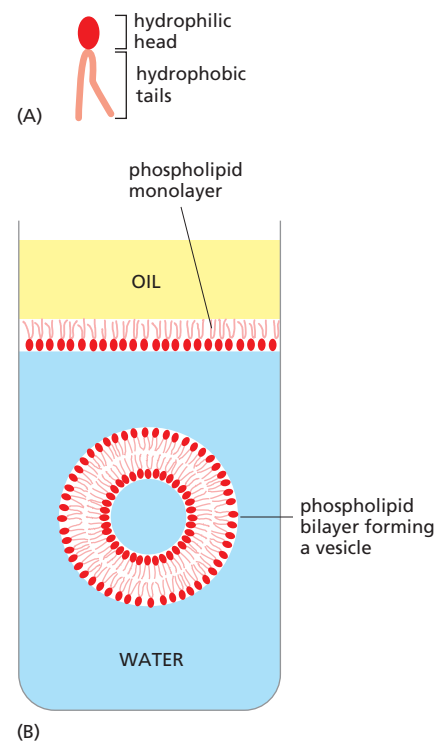


Figure 1–6 Behavior of phospholipid molecules in water. (A) A phospholipid molecule is amphiphilic, having a hydrophilic (water-loving) phosphate head group and a hydrophobic (water-avoiding) hydrocarbon tail. (B) At an interface between oil and water, phospholipids arrange themselves as a single sheet (a monolayer), with their head groups facing the water and their tail groups facing the oil. When immersed in water, however, phospholipids aggregate to form lipid bilayers that fold in on themselves to form sealed aqueous compartments known as vesicles.

Although the chemical details vary, the hydrophobic tails of the predominant lipid molecules in all cells are hydrocarbon polymers ($-\text{CH}_2-\text{CH}_2-\text{CH}_2-$), and their spontaneous assembly into a lipid bilayer is but one of many examples of an important general principle: cells produce molecules whose chemical properties cause them to *self-assemble* into the structures that a cell needs.

The cell boundary cannot be totally impermeable. If a cell is to grow and reproduce, it must be able to import raw materials and export waste across its plasma membrane. All cells therefore have specialized proteins embedded in their plasma membrane that transport specific molecules from one side to the other. Some of these *membrane transport proteins*, like some of the proteins that catalyze the fundamental small-molecule reactions inside the cell, have been so well conserved over the course of evolution that we can recognize the family resemblances between them when even the most distantly related organisms are compared.

The transport proteins in the plasma membrane largely determine which molecules enter the cell, while the catalytic proteins (enzymes) inside the cell determine the reactions that the entering molecules undergo. Thus, by specifying the RNAs and proteins that the cell produces, the genetic information recorded in the DNA sequence dictates the entire chemistry of the cell—in fact, not only its chemistry but also its form and its behavior, for these too are chiefly determined and controlled by the cell's proteins.

Cells Operate at a Microscopic Scale Dominated by Random Thermal Motion

Thus far we have described the cell as a self-replicating, membrane-bound bag of chemicals and macromolecules; but, as the unit of life, the cell is much more than the sum of its parts. Although not obvious from microscopy images, even the simplest cell is highly ordered internally: its individual components must self-assemble and become highly organized for the cell to function. And the cell contents are in perpetual motion. The most obvious movements are catalyzed by *motor proteins*, enzymes that use the energy of ATP hydrolysis for a wide variety of purposes; these include pumping ions across the plasma membrane, translocating large assemblies from one intracellular site to another, and propelling the cell through its environment. In addition, and as previously mentioned, random thermal motions of molecules (including water) are prominent at the scale of cells—whose dimensions can be as small as a micrometer (10^{-6} meters) in diameter. This type of spontaneous movement, called *thermal* or **Brownian motion**, was first observed by Robert Brown in 1827, while looking through a microscope at pollen grains immersed in water. Caused by random molecular collisions, the constant fluctuating movement has important repercussions. Brownian motion drives a process called *diffusion*, and it determines the rates of biochemical reactions as molecules collide with one another within the interior of a cell (described in Chapter 2; see Movie 2.4).

Even though random, the cell can harness Brownian motion for its own advantage. For example, during one step in the crawling migration of animal cells, the plasma membrane at the leading edge extends forward (see Chapter 16). This movement does not involve motor proteins. Instead, a cytoskeletal filament (an actin polymer) polymerizes adjacent to the inner membrane surface. When the membrane fluctuates in the forward direction, actin quickly fills in the gap so that the membrane cannot slip back to its original position. This phenomenon, in which random thermal motions are harnessed in a directed way, creates a *Brownian ratchet* (Figure 1-7).

Because an object at the micrometer scale is constantly buffeted by water molecules, its movement requires overcoming high viscous drag forces. As a result, the directed movement of a complex of molecules inside the cell (by a motor protein, for example) will stop immediately when the motor disengages, leaving the complex to be randomly buffeted by thermal motion. There is no “gliding” inside the cell.

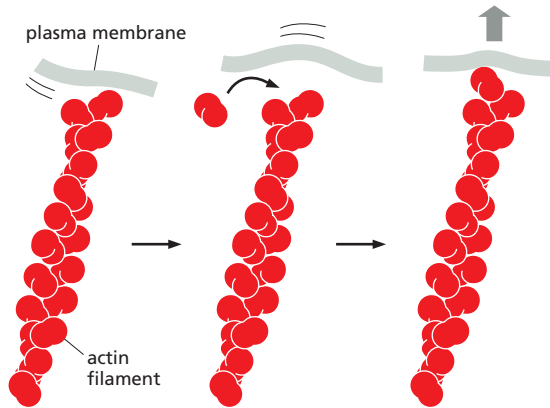


Figure 1–7 How membrane protrusion is driven by a simple Brownian ratchet.

A single actin filament is shown abutting the plasma membrane, which is fluctuating back and forth because of random thermal motions. When the membrane happens to move away from the end of the filament, it creates sufficient space for an additional subunit, which quickly adds on. The slightly longer filament acts as a ratchet and prevents the membrane from moving back to its original position. In a migrating animal cell, this Brownian ratchet process drives protrusion of the membrane and contributes to forward movement of the cell, as described in Chapter 16 (see pp. 956–957).

A Living Cell Can Exist with 500 Genes

We have seen how genomes carry the information for all the proteins and RNA molecules of a cell, and how, through catalysis, all the other building blocks of the cell are made. But how complex are real living cells? In particular, what are the minimum requirements of a living cell? One measure of complexity is based on the total number of genes in an organism's genome. A species that has one of the smallest known genomes is the bacterium *Mycoplasma genitalium*, which causes a common, sexually transmitted, human disease (Figure 1–8). This organism lives as a parasite in mammals, where the environment provides it with many of the small molecules it needs ready-made. Nevertheless, it still has to make all the large molecules—DNA, RNAs, and proteins. It has 525 genes, most of which are essential. Its genome of 580,070 nucleotide pairs represents 145,018 bytes of information—about as much as it takes to record the text of one chapter of this book. Cell biology may be complicated, but it is not unimaginably so.

Summary

The individual cell is the minimal self-reproducing unit of life. A cell consists of a self-replicating collection of catalysts, enclosed in a plasma membrane. All cells operate as biochemical factories, driven by the free energy released in a complicated network of chemical reactions. Central to a cell's ability to reproduce is the transmission of its genetic information to its progeny cells when it divides. All cells store their genetic information in double-strand DNA, and the complete sequence of DNA nucleotides for each organism is known as its genome. The cell replicates this information by separating the paired DNA strands and using each as a template for polymerization to make a new DNA strand with a complementary sequence of nucleotide subunits. The same strategy of templated polymerization is used in the transcription of portions of the DNA into molecules of the closely related polynucleotide polymer, RNA. Most of these RNA molecules are mRNAs that in turn guide the synthesis of protein molecules by the process of translation. Proteins are polymers of amino acid subunits and are the catalysts for almost all the cell's chemical reactions. They are also responsible for the selective import and export of molecules across the plasma membrane that surrounds each cell. The specific shape and function of each protein depend on its amino acid sequence, which is specified by the nucleotide sequence of a corresponding segment of the DNA—the gene that codes for that protein. In this way, the DNA of the cell determines the cell's chemistry, which is fundamentally similar in all cells, reflecting their ultimate origin from a common ancestor cell that existed on Earth more than 3.5 billion years ago.

GENOME DIVERSIFICATION AND THE TREE OF LIFE

The success of living organisms based on DNA, RNA, and protein has been spectacular. Through its billions of years of proliferation, life has populated the oceans, covered the land, penetrated deep into Earth's crust, and molded the surface of

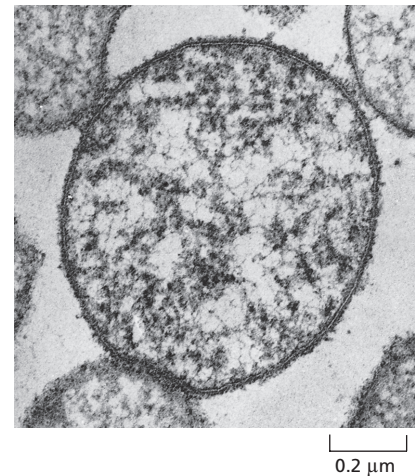


Figure 1–8 The small bacterium *Mycoplasma genitalium*. It is viewed here in cross section in an electron microscope, which uses a beam of electrons instead of light to create an image with a resolution that is many times higher than that of an image viewed in a conventional light microscope. Of the 525 genes this bacterium contains, 43 code for transfer RNAs, ribosomal RNAs, and other nonprotein-coding RNAs. Of the 482 protein-coding genes, 154 are involved in replication, transcription, translation, and related processes involving DNA, RNA, and protein; 98 are involved in the membrane and surface structures of the cell; 46 are involved in the transport of nutrients and other molecules across the plasma membrane; and 71 are involved in energy conversion and the synthesis and degradation of small molecules. (Courtesy of Roger Cole, in *Medical Microbiology*, 4th ed. [S. Baron, ed.]. Galveston: University of Texas Medical Branch, 1996.)

our planet. Our oxygen-rich atmosphere, the deposits of coal and oil, the layers of iron ores, the cliffs of chalk and limestone and marble—all these are products, directly or indirectly, of past biological activity on Earth.

Living things are not confined to the familiar temperate realm of land, water, and sunlight inhabited by plants and animals. They are found in the darkest depths of the ocean, in hot volcanic mud, in pools beneath the frozen surface of the Antarctic, and buried kilometers deep in Earth's crust. The creatures that live in these extreme environments are generally unfamiliar, not only because they are inaccessible, but also because they are mostly microscopic and cannot be maintained in a laboratory. Even in more familiar habitats, most organisms are too small for us to see without special equipment: they tend to go unnoticed, unless they cause a disease or rot the timbers of our houses. Yet such *microorganisms* (microbes) are by far the most numerous living organisms on our planet. Only recently, through new methods of molecular analysis including rapid *DNA sequencing*, have we begun to get a picture of life on Earth that is not grossly distorted by our biased perspective as large animals living on dry land.

In this section, we consider—in very broad terms—the diversity of organisms on our planet and the relationships among them. Because the genetic information for every organism is written in the universal language of DNA sequences, and because the DNA sequence of any organism's genome can be readily determined, it is now possible to characterize, catalog, and compare any set of living organisms with reference to these sequences. From such comparisons we can specify the place of each organism in the family tree of living species—the “tree of life.”

The Tree of Life Has Three Major Domains: Eukaryotes, Bacteria, and Archaea

The classification of living things has traditionally depended on comparisons of their outward appearances: we can see that a fish has eyes, jaws, backbone, brain, and so on, just as humans do, and that a worm does not—just as we can see that a rosebush is more similar to an apple tree than to grass. As Darwin showed, we can readily interpret such close family resemblances in terms of an evolution from common ancestors, and we can find the remains of many of these ancestors preserved in the fossil record. In this way, it became possible to draw a family tree of living organisms, showing the various lines of descent, as well as branch points in evolutionary history, where the ancestors of one group of species became different from those of another.

When the disparities between organisms become very great, however, these methods begin to fail. How do we decide whether a fungus is more closely related to a plant or to an animal? When it comes to microscopic organisms such as bacteria, the task becomes harder still: one tiny rod or sphere can look much like another. Moreover, much of our knowledge of the microbial world was traditionally restricted to those species that can be isolated and cultured in the laboratory. But direct DNA sequencing of populations of microbes in their natural habitats—such as soil, ocean water, or even the human mouth—has taught us that the vast majority of microbes cannot be easily cultured in the laboratory. Often, they thrive in the wild as components of complex ecosystems and—when separated from their natural surroundings—cannot survive. Until modern DNA sequencing was developed, these organisms were largely unknown to us, especially those that inhabit extreme environments such as the deep Earth's crust or seawater miles below the ocean surface.

Genome analysis has now provided us with a simple, direct, and powerful way to determine evolutionary relationships. The complete DNA sequence of an organism defines its nature with almost perfect precision and in exhaustive detail. Moreover, this specification is in a digital form—a string of letters—that can be entered into a computer and compared with the corresponding information for any other organism. Because DNA is subject to random changes that accumulate over long periods of time (as we will see shortly), the number of differences

between the DNA sequences of two organisms can provide a direct, objective, and quantitative indication of the evolutionary distance between them.

For constructing a comprehensive tree of life, it is necessary to begin with a segment of DNA that is easily recognized in the genomes of all organisms. We discussed earlier how all cells use the same fundamental mechanism to translate a nucleotide sequence into a protein sequence, and we saw that the ribosome is the “decoding machine” that carries this out. Ribosomes are fundamentally similar in all organisms, and an especially well-conserved component of them is the RNA molecules that make up their core. Although the exact sequence of these *ribosomal RNAs* (*rRNAs*) differs across organisms, they are similar enough to use them as a ruler to judge how closely two species are related: the more similar the ribosomal RNA sequences, the more recently the two species diverged from a common ancestor and the more related they must be. Once a rough approximation of the tree of life has been obtained in this way, many additional DNA sequences in genomes—those that might not be identifiable in all organisms—can accurately determine relationships among more closely related species.

This approach has revealed that the living world consists of three major divisions, or *domains*: eukaryotes, bacteria, and archaea, as illustrated in Figure 1–9; in the following paragraphs, we briefly introduce each in turn.

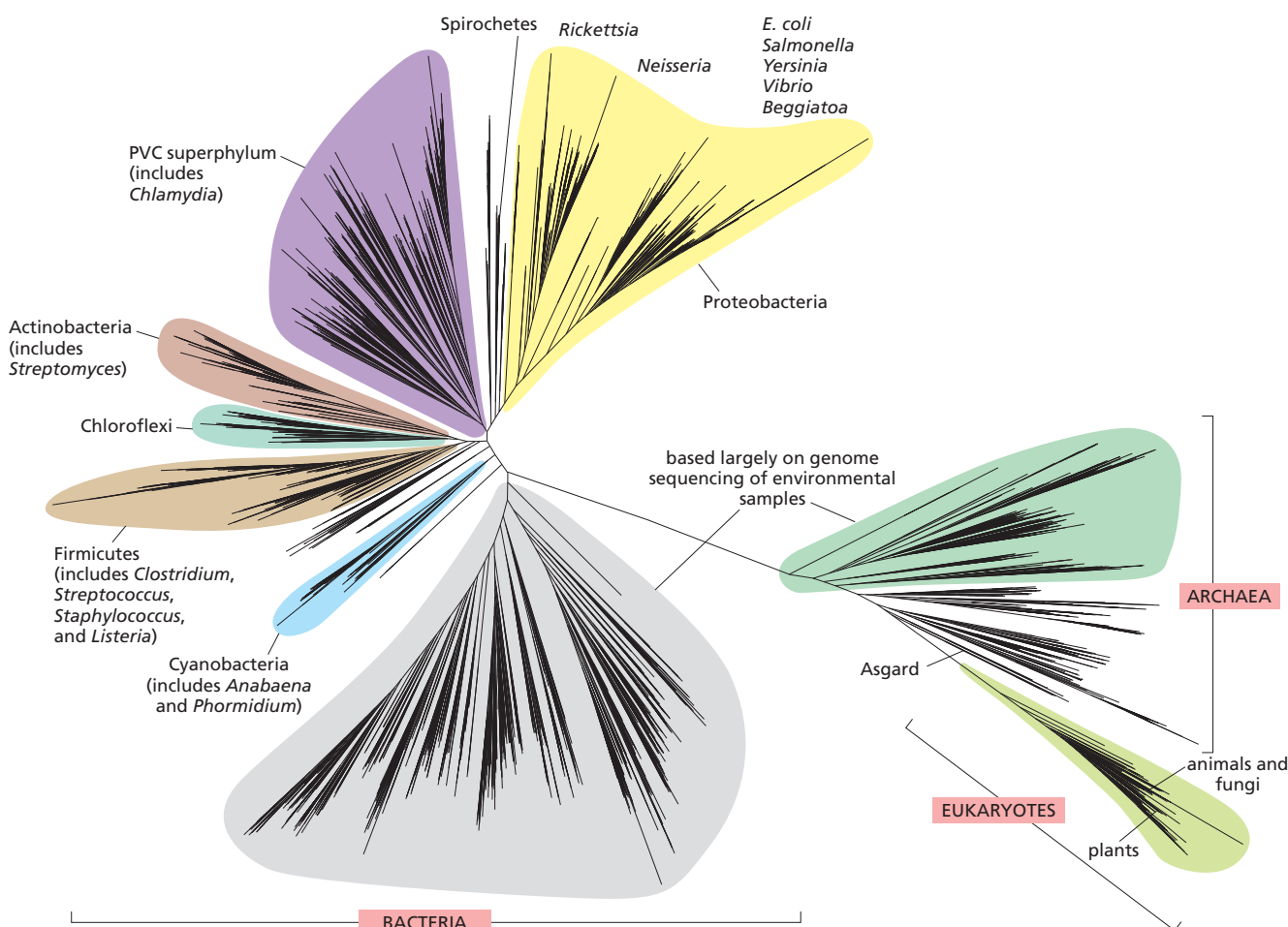


Figure 1–9 A global tree of life, based on genome comparisons, shows the three major divisions (domains) of the living world. The lengths of the branches are proportional to differences among genomes using common genes that can be recognized and compared across many different species. Some of the organisms discussed in this and later chapters are indicated. Of the three domains of life (bacteria, archaea, and eukaryotes), bacteria encompass by far the greatest diversity, commensurate with their ability to colonize nearly every ecological niche on the planet. So many new bacterial species are currently being identified through DNA sequencing of environmental samples that simply naming them has become a challenge. Although eukaryotes (and especially animals) are the main focus of this book, they comprise only a small slice of the global diversity. An expanded eukaryotic tree is shown in Figure 1–35, and an expanded tree of mammals is given in Figure 4–67. (Adapted from C.J. Castelle and J.F. Banfield, *Cell* 172:1181–1197, 2018.)

Eukaryotes Make Up the Domain of Life That Is Most Familiar to Us

The great variety of living creatures that we see around us are eukaryotes. The name is from the Greek, meaning “truly nucleated” (from the words *eu*, “well” or “truly,” and *karyon*, “kernel” or “nucleus”), reflecting the fact that the cells of these organisms have their DNA enclosed in a membrane-bound organelle called the *nucleus*. Visible by simple light microscopy, this feature was used in the early twentieth century to classify living organisms as either **eukaryotes** (those with a nucleus) or **prokaryotes** (those without a nucleus). We now know that prokaryotes comprise two of the three major domains of life, the bacteria and archaea. Eukaryotic cells are typically much larger than those of bacteria and archaea; in addition to a nucleus, they typically contain a variety of membrane-bound organelles that are also lacking in the prokaryotes. The genomes of eukaryotes also tend to run much larger—containing more than 20,000 genes for humans and corals, for example, compared with 4000–6000 genes for the typical bacteria or archaea.

In addition to plants and animals, the eukaryotes include fungi (such as mushrooms or the yeasts used in beer- and bread-making), as well as an astonishing variety of single-celled, microscopic forms of life. Most of this book is focused on the cell biology of eukaryotic organisms (especially animals); in the final sections of this chapter, we shall return to eukaryotes and focus on the variety within this group.

On the Basis of Genome Analysis, Bacteria Are the Most Diverse Group of Organisms on the Planet

When modern trees of life were constructed using genome information, one of the big surprises was how much more evolutionarily diverse the bacterial world is compared with the eukaryotes; we now know that this great diversity reflects the much earlier appearance of bacteria in the evolutionary history of the planet. Bacteria are usually very small (and invisible to the unaided eye), and they generally live as independent individuals or in loosely organized communities, rather than as multicellular organisms. They are typically spherical or rod-shaped and measure a few micrometers (μm) in linear dimension (Figure 1–10). They often have a tough protective coat, called a cell wall, beneath which a plasma membrane encloses a single cytoplasmic compartment—the *cytoplasm*—containing DNA, RNA, proteins, and the many small molecules needed for life (Figure 1–11). Although difficult to discern in the light microscope, the interior of a bacterium is nevertheless highly organized, a topic we discuss in Chapter 16.

Commensurate with the diversity of their genomes, bacteria live in an enormous variety of ecological niches, and they are astonishingly varied in their

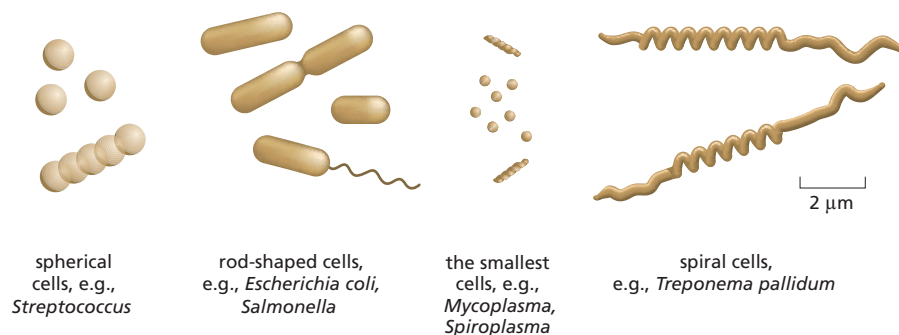


Figure 1–10 Shapes and sizes of some bacteria. Although most are small, as shown, measuring a few micrometers in linear dimension, there are also some giant species. An extreme example is the cigar-shaped bacterium *Epulopiscium fishelsoni*, which lives in the gut of a surgeonfish and can be up to 600 μm long (not shown).

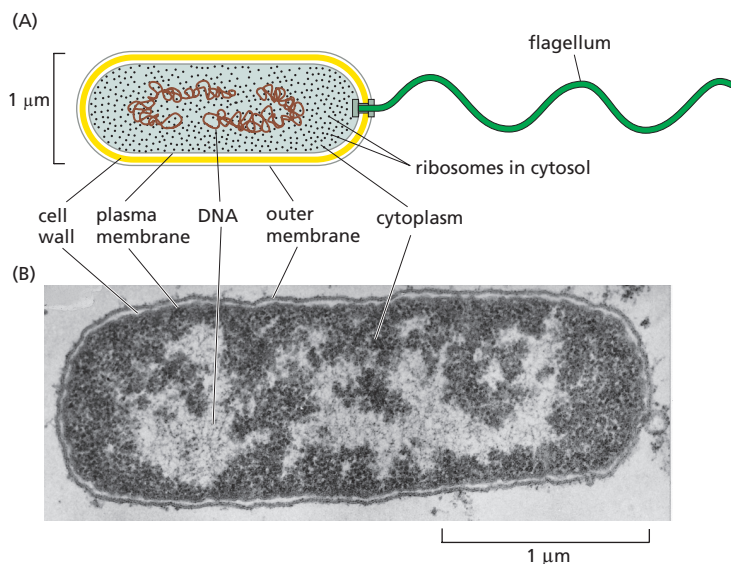


Figure 1-11 Bacterial structure. (A) A drawing of the bacterium *Vibrio cholerae*, showing its simple internal organization. This species can infect the human small intestine to cause cholera; the severe diarrhea that accompanies this disease kills more than 100,000 people a year worldwide. Like many other bacteria, *Vibrio* has a helical appendage at one end—a flagellum—that rotates as a propeller to drive the cell forward. (B) An electron micrograph of a longitudinal section through the widely studied bacterium *Escherichia coli* (*E. coli*). *E. coli* is part of our normal intestinal *microbiota*, the complete collection of microbes in our gut. It has many flagella distributed over its surface, but they are not visible in this section. Both of the bacteria shown here are Gram negative, having both an outer and an inner (plasma) membrane. However, many bacterial species lack the outer membrane; these are classified as Gram positive. (B, courtesy of E. Kellenberger.)

biochemical capabilities. There exist species that can utilize virtually any type of organic molecule as food, ranging from sugars and amino acids to hydrocarbons, including the simplest hydrocarbon, methane gas (CH_4). Other species (Figure 1-12) harvest light energy in a variety of ways; some, like plants, carry out photosynthesis and generate oxygen as a by-product. Still others can feed on a plain diet of inorganic nutrients, getting their carbon from CO_2 , and relying on a host of other chemicals that occur in the environment to fuel their energy needs—including H_2 , Fe^{2+} , H_2S , and elemental sulfur (Figure 1-13).

A wide range of bacteria directly affect human health. The bubonic plague of the Middle Ages (estimated to have killed half the population of Europe) and the current tuberculosis pandemic (more than a million deaths a year) are each due to a specific species of bacteria. And thousands of different bacterial species reside in our gut and on our skin, where they are often beneficial to us. We shall discuss bacteria throughout the book, as it is the study of these relatively simple cells that led to much of our understanding of basic biological processes—including DNA replication, transcription, and translation. We focus again on bacteria in Chapter 24 when we examine the cell biology of infectious disease. Finally, genetic

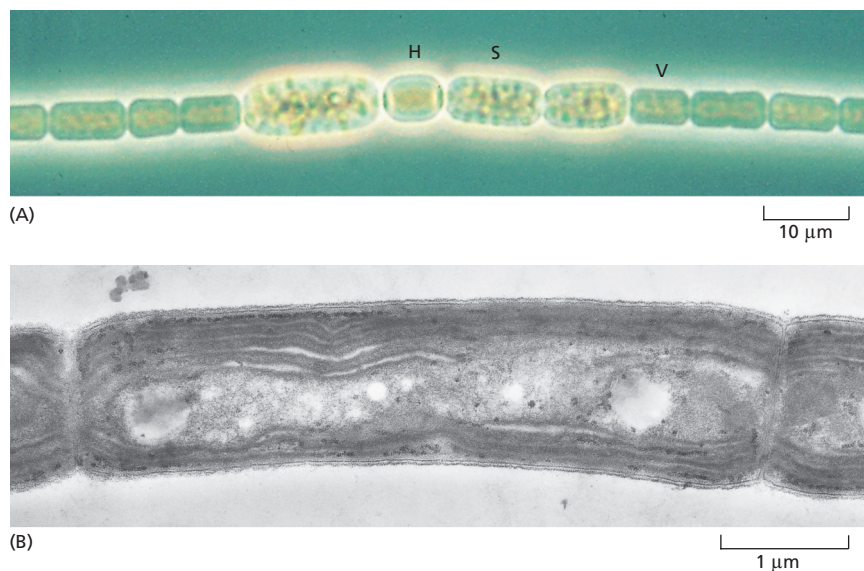


Figure 1-12 Photosynthetic bacteria. (A) A light micrograph of the bacterium *Anabaena cylindrica*. Its cells form long chains, in which most of the cells (labeled V) perform photosynthesis (and thereby capture CO_2 and incorporate C into organic compounds); others (labeled H) become specialized for fixing N from N_2 ; and still others (labeled S) develop into spores, which can resist unfavorable conditions. (B) An electron micrograph of a related photosynthetic bacterium, *Phormidium lamosum*, which shows the intracellular membranes where photosynthesis occurs. As shown in these micrographs, some prokaryotes have intracellular membranes and form colonies that resemble simple multicellular organisms. (A, courtesy of David Adams; B, courtesy of D.P. Hill and C.J. Howe.)

engineering techniques allow bacteria to be put to use as small “factories” to produce human pharmaceuticals, biofuels, and other high-value chemical products, as we discuss in Chapter 8.

Archaea: The Most Mysterious Domain of Life

Of the three domains of life, archaea remains the most poorly understood. Most of its members have been identified only by DNA sequencing of samples from the environment, and relatively few have been cultured and studied up close in the laboratory. Like bacteria, the archaea we know most about are small and lack the internal, membrane-bound organelles that distinguish the eukaryotes. But they differ from bacteria in many ways, including the chemistry of their cell walls, the kinds of lipids that make up their membrane, and the range of biochemical reactions that they can carry out. Another surprising conclusion came from genome comparisons: although archaea resemble bacteria in their outward appearances, their genomes are much more closely related to eukaryotes than to bacteria (see Figure 1–9). It has even been proposed that the tree of life should be considered to have only two principal domains, with the archaea and eukaryotes making up one domain and bacteria constituting the other. The close relationship of archaea and eukaryotes has also changed our views on how the earliest eukaryotic cell evolved, a topic addressed later in this chapter.

At first it was thought that archaea occupied only extreme environments such as volcanoes, salt lakes, acid hot springs, and the stomachs of cattle, but they are now recognized to be present also in more congenial surroundings such as soils, seawater, and our skin. Commensurate with the wide variety of ecological niches in which they have been found, different species of archaea have highly diverse chemistries. They are believed to be the predominant life-form in soil and seawater, and they play major roles in recycling nitrogen and carbon, two of the most important elements for all cells.

Organisms Occupy Most of Our Planet

To understand life on Earth, we need to understand more than its diversity; we also need to know where life is found on our planet and how various living species are distributed. Organisms inhabit nearly all of the planet, and we continue to discover new habitats. Amazingly, some bacteria and archaea even live miles down in Earth’s deep crust and in the deepest and most hostile parts of the oceans.

How are the main groups of organisms distributed among different environments? DNA sequencing and other advanced technologies have been used recently to address this question. The total biomass on Earth is estimated to contain ~550 gigatons (10^{15} grams) of carbon, of which 450 gigatons of carbon (Gt C) is plants, 70 Gt C is bacteria, 7 Gt C is archaea, and 2 Gt C is animals (Figure 1–14). The plants are mainly terrestrial; the bacteria and archaea are mainly in the soil and Earth’s crust. Total terrestrial biomass is 100 times greater than that in the oceans, although most of the animal mass is found in the oceans. The human biomass is 10 times greater than that of all measurable wild animals together, and—while human biomass continues to increase—that of wild animals is falling, largely as a result of human activities.

Although humans and other animals make up a small fraction of Earth’s biomass, their existence depends completely on other forms of life. In the next section, we shall see some of the ways that these different life-forms work together to capture and recycle energy from Earth’s inanimate features.

Cells Can Be Powered by a Wide Variety of Free-Energy Sources

Organisms obtain the free energy needed for life in different ways. Some—such as animals, fungi, and the many different bacteria that live in the human gut—get it by feeding on other living things or the organic chemicals they produce; such organisms are called *organotrophic* (from the Greek word *trophe*, meaning



Figure 1–13 The bacterium *Beggiatoa*. It lives in sulfurous environments (for example, see Figure 1–15) and gets its energy by oxidizing H_2S ; it can fix carbon even in the dark. Note the yellow deposits of sulfur inside the cells. (Courtesy of Ralph S. Wolfe.)

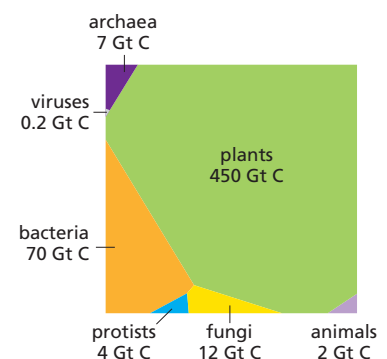


Figure 1–14 The distribution of living biomass on Earth. The total biomass on Earth expressed as gigatons of carbon (Gt C) is estimated to be ~550 Gt C. In the graph shown, the area of each taxon represented is proportional to the taxon’s global biomass, so plants account for about 80% (450/550) of the total biomass, whereas animals account for 0.4% (2/550). These recent estimates are based on various advanced techniques, including DNA sequencing and remote sensing. (Adapted from Y.M. Bar-On et al., *Proc. Natl. Acad. Sci. USA* 115:6506–6511, 2018. With permission from the authors.)

“food”). Others derive their free energy directly from the nonliving world. These primary energy converters fall into two classes: those that harvest the energy of sunlight, and those that capture their energy from energy-rich systems of inorganic chemicals in the environment (chemical systems that are far from chemical equilibrium). Organisms of the former class are called *phototrophic* (feeding on sunlight); those of the latter are called *lithotrophic* (feeding on rock). The organo-organisms like ourselves could not exist without these primary energy converters, which are the most plentiful form of life.

The phototrophic organisms include many types of bacteria, as well as algae and plants, on which we—and virtually all the living things that we ordinarily see around us—depend. Phototrophic organisms have changed the whole chemistry of our environment: as a prime example, the oxygen in Earth’s atmosphere is a by-product of their biosynthetic activities.

Lithotrophic organisms are not such an obvious feature of our world, because they are microscopic and mostly live in habitats that humans do not frequent—deep in the ocean, buried in Earth’s crust, or in various other seemingly inhospitable environments. But they are a major part of the living world, and they are especially important in any consideration of the history of life on Earth.

Some lithotrophs get energy from *aerobic* reactions, which use molecular oxygen from the environment; because atmospheric O_2 is ultimately the product of living phototrophic organisms, these aerobic lithotrophs are, in a sense, feeding on the products of past life. There are, however, many other lithotrophs that live *anaerobically*, in places where little or no molecular oxygen is present; these are circumstances similar to those that existed in the early days of life on Earth, before oxygen had accumulated.

The most dramatic of the anaerobic sites are the hot *hydrothermal vents* on the floor of the Pacific and Atlantic Oceans. They are located where the ocean floor is spreading as new portions of Earth’s crust form by a gradual upwelling of material from Earth’s interior (Figure 1–15). Downward-percolating seawater is heated and driven back upward as a submarine geyser, carrying with it a current of chemicals from the hot rocks below. A typical cocktail might include H_2S , H_2 , CO , Mn^{2+} , Fe^{2+} , Ni^{2+} , CH_4 , NH_4^+ , and phosphorus-containing compounds.

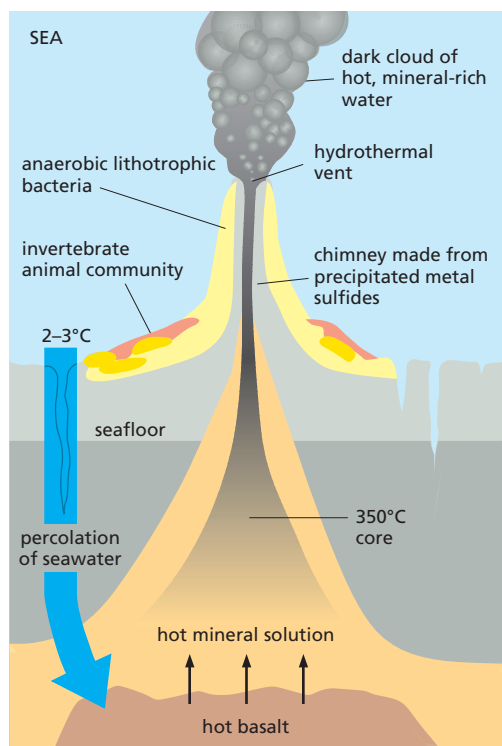


Figure 1–15 The geology of a hot hydrothermal vent in the ocean floor.

As indicated, seawater percolates down toward the hot, molten, volcanic rock upwelling (basalt) from Earth’s interior and is heated and driven back upward, carrying a mixture of minerals leached from the hot rock. A temperature gradient is set up, from more than $350^{\circ}C$ near the core of the vent, down to $2\text{--}3^{\circ}C$ in the surrounding ocean. Minerals precipitate from the water as it cools, forming a chimney. Different classes of organisms, thriving at different temperatures, live in different neighborhoods of the chimney. A typical chimney might be a few meters tall, spewing out hot, mineral-rich water. The locations of lithotrophic bacteria and the invertebrate marine animals that depend on them are also shown (see Figure 1–16).

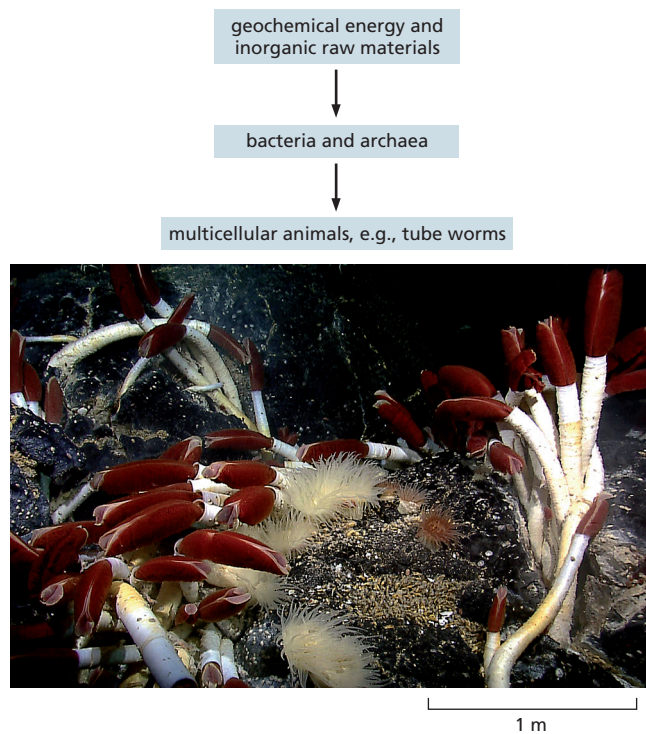


Figure 1–16 Organisms living at a depth of 2500 meters near a vent in the ocean floor. Close to the vent, at temperatures up to about 120°C, various lithotrophic species of bacteria and archaea live, directly fueled by geochemical energy. A little further away, where the temperature is lower, various invertebrate animals live by feeding on these microorganisms. Most remarkable are the giant (2-meter-long) tube worms, *Riftia pachyptila*, which are shown in the photograph. Rather than feed on the lithotrophic microbes, these worms live in symbiosis with them: specialized organs in the worms harbor huge numbers of symbiotic sulfur-oxidizing bacteria, which harness geochemical energy and supply nourishment to their hosts, which have no mouth, gut, or anus. The tube worms are thought to have evolved from more conventional animals and to have become secondarily adapted to life at hydrothermal vents. (Science History Images/Alamy Stock Photo.)

A dense population of microorganisms lives in the neighborhood of the vent, thriving on this austere diet and harvesting free energy from reactions between the available chemicals. Various invertebrate marine animals—clams, mussels, and giant marine worms—in turn, live off the microbes at the vent, forming an entire ecosystem analogous to the world of plants and animals that we belong to, but one powered by geochemical energy instead of light (**Figure 1–16**).

Some Cells Fix Nitrogen and Carbon Dioxide for Other Cells

To make a living cell requires matter, as well as free energy. DNA, RNA, and protein are composed of just six elements: hydrogen, carbon, nitrogen, oxygen, sulfur, and phosphorus. These are all plentiful in the nonliving environment, in Earth's rocks, water, and atmosphere. But they are not present in chemical forms that allow easy incorporation into biological molecules. Atmospheric N_2 and CO_2 , in particular, are extremely unreactive. A large amount of free energy is required to drive the reactions that use these inorganic molecules to make the organic compounds needed for further biosynthesis; that is, to *fix* nitrogen and carbon dioxide, so as to make N and C available to living organisms. Many types of cells lack the biochemical machinery to achieve this fixation; they instead rely on other classes of cells to do the job for them. We animals depend on plants, directly or indirectly, for our supplies of carbon- and nitrogen-containing organic compounds. Plants in turn, although they can fix carbon dioxide from the atmosphere, lack the ability to fix atmospheric nitrogen; they depend in part on nitrogen-fixing bacteria to supply their need for nitrogen-containing organic compounds. Plants of the pea family, for example, harbor symbiotic nitrogen-fixing bacteria in nodules in their roots.

Because living cells can differ widely in some of the most basic aspects of their biochemistry, cells with complementary needs and capabilities have frequently developed close associations. Some of these *symbiotic* associations, as we will see later, have evolved to the point where the partners have lost their separate identities altogether: they have joined forces to form a single composite cell—an *endosymbiotic* association, as opposed to an *ectosymbiotic* one between separate organisms.

Genomes Diversify Over Evolutionary Time, Producing New Types of Organisms

Having discussed our current views on the diversity of life-forms, how they are distributed across Earth, and how they depend on one another, we now turn to the question of how this great diversity was generated. All life depends on the storage of genetic information in the form of each organism's DNA genome, so our focus is on how genomes change over evolutionary time.

In storing and copying genetic information, random accidents and errors occur, altering the nucleotide sequence; that is, creating **mutations**. Therefore, when a cell divides, the genomes of its two daughters are often not quite identical to each other or to that of the parent cell. On rare occasions, the error may represent a change for the better; more probably, it will cause no significant difference in the cell's prospects. But in some cases, the error will cause serious damage; for example, by disrupting the coding sequence for a key protein or RNA molecule. Changes due to mistakes of the first type will tend to be perpetuated, because the altered cell has an increased likelihood of surviving and reproducing itself. Changes due to mistakes of the second type—*neutral* changes—may be perpetuated or not: in the competition for limited resources, it is a matter of chance whether the altered cell or its cousins will succeed. But changes that cause serious damage lead nowhere: the cell that suffers them dies, leaving no progeny. Through endless repetition of this cycle of error and trial—of *mutation* and *natural selection*—organisms evolve: their genetic specifications change, sometimes giving organisms new ways to exploit the environment more effectively, to survive in competition with others, and to reproduce successfully.

Some parts of the genome will change more readily than others in the course of evolution. A segment of DNA that does not code for protein or RNA and has no significant regulatory role is free to change at a rate limited only by the frequency of random errors. In contrast, a gene that codes for a highly optimized, essential protein or RNA molecule cannot alter so easily: when mistakes occur, the faulty cells are almost always disabled and eliminated. Genes of this latter sort are therefore *highly conserved*. Through 3.5 billion years or more of evolutionary history, many DNA sequences have changed beyond all recognition, but the most highly conserved genes remain perfectly recognizable in all living species.

These latter genes are the ones we must examine if we wish to trace family relationships between the most distantly related organisms in the tree of life. We discussed an example of one such gene—that for ribosomal RNA—when we introduced the classification of the living world into the three domains of eukaryotes, bacteria, and archaea. Because the production of proteins is fundamental to all living cells, this component of the ribosome has been highly conserved since early in the history of life on Earth (Figure 1–17).

The ribosomal RNA genes are exceptional in being so well conserved, whereas most parts of genomes have diversified much more dramatically over evolutionary time. A complete DNA sequence for an organism—its genome sequence—reveals all the genes that an organism possesses, as well as those it lacks. When we

GTTCGGGGGGAGTATGGTTGCAAAGCTGAACTTAAAGGAATTGACGGAAGGGCACCACCAGGAGTGGAGCCTGCGGCTTAATTTGACTCAACACGGGAAACCTCACCC	human
GCCGCCTGGGGAGTACGGTCGCAAGACTGAACTTAAAGGAATTGGCGGGGAGCACTACAACGGTGGAGCCTGCGGTTTAATTGGATTCAACGCCGGGCATCTTACCA	<i>Methanococcus</i>
ACCGCCTGGGGAGTACGGCCGCAAGGTTAAACTCAAATGAATTGACGGGGGCCCG•ACAAGCGTGGAGCATGTGGTTTAATTCGATGCAACGCGAAGAACCTTACCT	<i>E. coli</i>
GTTCGGGGGGAGTATGGTTGCAAAGCTGAACTTAAAGGAATTGACGGAAGGGCACCACCAGGAGTGGAGCCTGCGGCTTAATTTGACTCAACACGGGAAACCTCACCC	human

Figure 1–17 Genetic information conserved since the days of the last universal common ancestor of all living things.

A part of the gene that codes for the smaller of the two main ribosomal RNA (rRNA) molecules in the ribosome is shown. (The complete molecule is about 1500–1900 nucleotides long, depending on the species.) Corresponding segments of nucleotide sequences from an archaeon (*Methanococcus jannaschii*), a bacterium (*Escherichia coli*), and a eukaryote (*Homo sapiens*) are aligned. The *red vertical lines* indicate sites where the nucleotides are identical between the species; the human sequence is repeated at the bottom of the alignment so that all three two-way comparisons can be seen. The *black dot* halfway along the *E. coli* sequence denotes a site where a nucleotide has been either deleted from the bacterial lineage in the course of evolution or inserted in the other two lineages. Note that the sequences from these three organisms, representative of the three domains of the living world, still retain unmistakable similarities.

compare the three domains of the living world, we can begin to see which genes are common to all of them—and must therefore have been present in the last universal common ancestral cell that was the founder of all present-day living things. We can also identify those genes that are peculiar to a single branch in the tree of life. To explain such findings, we need to consider how new genes arise and, more generally, how genomes evolve.

New Genes Are Generated from Preexisting Genes

The raw material of evolution is the DNA sequence that already exists: there is no natural mechanism for making long stretches of new, random, DNA sequence. In this sense, no gene is ever entirely new. Innovation can, however, occur in several ways (**Figure 1–18**):

1. *Intragenic mutation*: an existing gene can be randomly modified by changes in its DNA sequence, through various types of errors that occur in the process of DNA replication and DNA repair.
2. *Gene duplication*: an existing gene can be accidentally duplicated, creating a pair of initially identical genes within a single cell; these two genes may then diverge in the course of evolution.
3. *DNA segment shuffling*: two or more existing genes can break and rejoin to make a hybrid gene consisting of DNA segments that originally belonged to separate genes.
4. *Horizontal (intercellular) DNA transfer*: a piece of DNA can be transferred from the genome of one cell to that of another—including between species. This process contrasts with the usual *vertical transfer* of genetic information from parent to progeny.

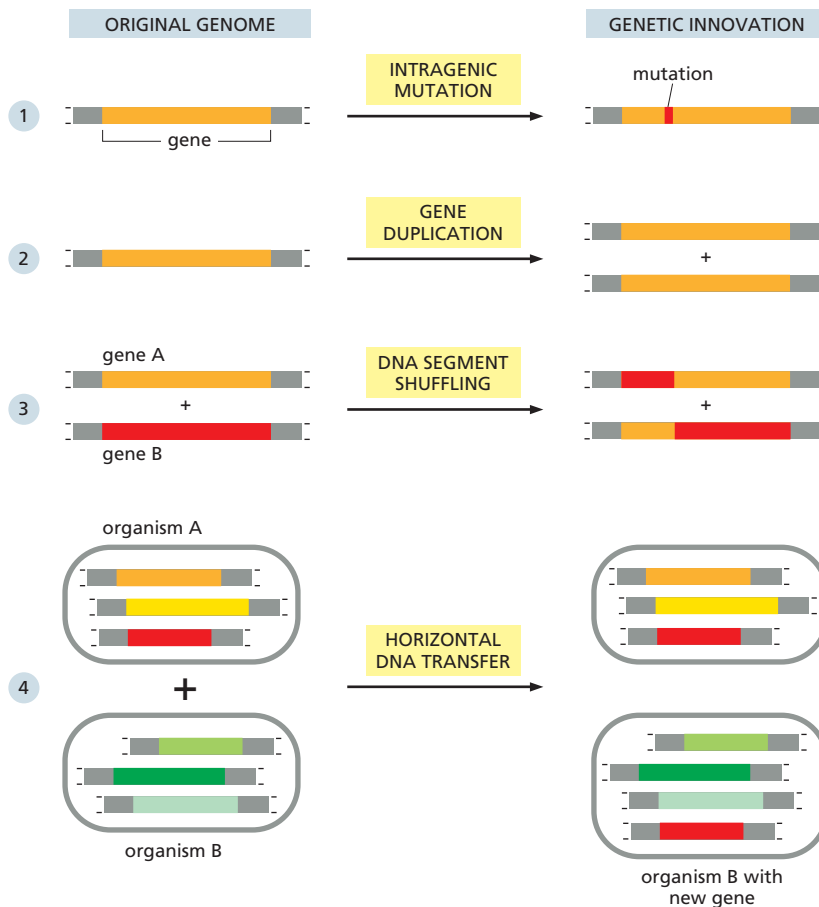


Figure 1–18 Four modes of genetic innovation and their effects on the DNA sequence of an organism. A special form of horizontal transfer occurs when cells of two different species enter into a permanent symbiotic association; genes from one of the cells may subsequently be transferred to the genome of the other, as we will see later when we discuss the likely evolutionary origins of mitochondria and chloroplasts.

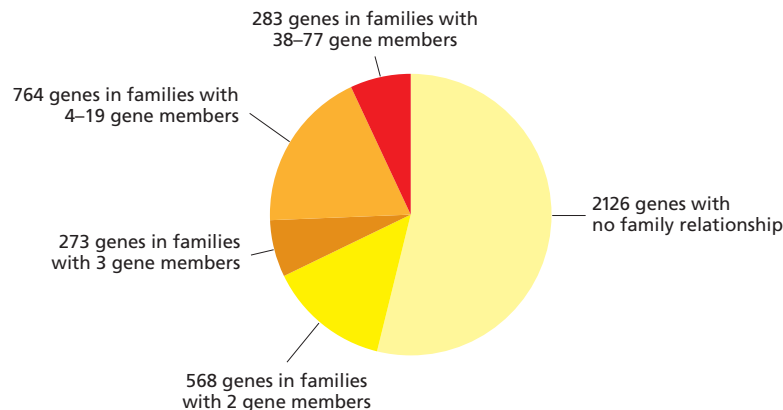


Figure 1–19 Families of evolutionarily related genes in the genome of *Bacillus subtilis*. The largest gene family in this bacterium consists of 77 genes coding for varieties of a class of membrane transport proteins called ABC transporters, which are found in all three domains of the living world. (Adapted from F. Kunst et al., *Nature* 390:249–256, 1997.)

Each of these types of change leaves a characteristic trace in the DNA sequence of the organism, and there is clear evidence that all four processes have occurred frequently during evolution. In Chapters 4 and 5, we discuss the mechanisms underlying these changes, but for the present we focus on the consequences.

Gene Duplications Give Rise to Families of Related Genes Within a Single Genome

A cell duplicates its entire genome each time it divides into two daughter cells. However, accidents occasionally result in the inappropriate duplication of just part of the genome, with retention of both the original and duplicate segments in a single cell. Once a gene has been duplicated in this way (see mode 2 in Figure 1–18), the two gene copies can acquire mutations and become specialized to perform different functions within the same cell and its descendants. Repeated rounds of this process of gene duplication and divergence, over many millions of years, have enabled one gene to give rise to a family of related genes within a single genome. Analysis of the DNA sequence of prokaryotic genomes reveals many examples of such **gene families**: in the bacterium *Bacillus subtilis*, for example, 47% of the genes have one or more obvious relatives (Figure 1–19).

The above evolutionary process must be distinguished from the genetic divergence that occurs when one species of organism splits into two separate lines of descent at a branch point in the family tree—when the human line separated from that of chimpanzees, for example. In the latter case, the genes gradually become different in the course of evolution, but they are likely to continue to have corresponding functions in the two sister species. Genes that are related by descent in this way—that is, genes in two separate species that derive from the same ancestral gene in the last common ancestor of those two species—are called **orthologs**. Related genes that have resulted from a gene duplication event within a single genome—and are likely to have diverged in their function—are called **paralogs**. Genes that are related by descent in either way are called **homologs**, a general term used to cover both types of relationship (Figure 1–20).

The Function of a Gene Can Often Be Deduced from Its Nucleotide Sequence

Family relationships among genes are important not just for their evolutionary interest, but also because they simplify the task of deciphering gene functions. Once the nucleotide sequence of a newly discovered gene has been determined, a scientist can tap a few keys on a computer to search large databases of known gene sequences for gene relatives. In many cases, the function of one or more of these homologs will have been already determined experimentally—generally in one of the model organisms described later in this chapter. Because gene sequence determines gene function, one can frequently make a good guess at the new gene’s function, as it is likely to be similar to that of the already

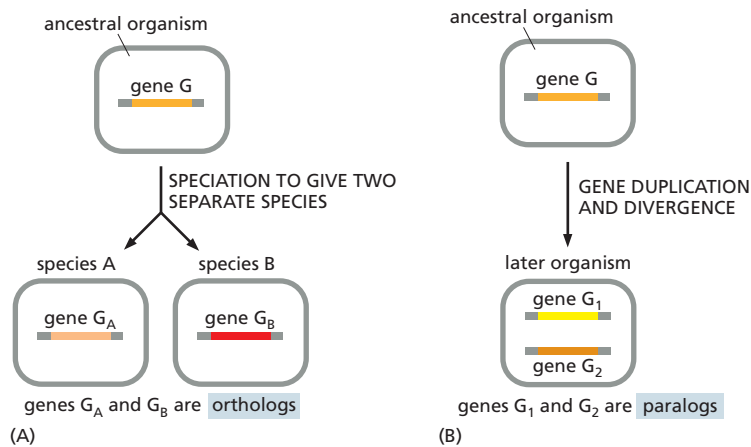


Figure 1-20 Two types of gene homology based on different evolutionary pathways. (A) Orthologs. (B) Paralogs. Genes related by either mechanism are called homologs.

known homologs. In this way, it is possible to decipher a great deal about the biology of an organism simply by analyzing the DNA sequence of its genome.

More Than 200 Gene Families Are Common to All Three Domains of Life

Given the complete genome sequences of representative organisms from all three domains of life—eukaryotes, bacteria, and archaea—we can search systematically for homologies that span this enormous evolutionary divide. In this way, we can begin to take stock of the common inheritance of all living things. There are considerable difficulties in this enterprise. For example, individual species have often lost some of the ancestral genes, and other genes have almost certainly been acquired by horizontal transfer from another species and therefore are not truly ancestral. In fact, genome comparisons strongly suggest that both lineage-specific gene loss and horizontal gene transfer, in some cases between evolutionarily distant species, have been major factors in evolution, at least among bacteria and archaea. As an additional difficulty, in the course of 2 or 3 billion years, some genes that were initially shared will have changed beyond recognition through mutation.

Because of all these vagaries of the evolutionary process, it is difficult, if not impossible, to determine the ancestral gene set that diversified into the present-day variety of life. A crude approximation can be obtained by tallying the gene families that have representatives in multiple—but not necessarily all—species from the three major domains of life. One such analysis revealed 264 ancient conserved families, each of which could be assigned a function on the basis of the best-characterized family member. As shown in [Table 1-1](#), the largest number of shared gene families were involved in translation and in amino acid metabolism and transport. However, it must be emphasized that this set of highly conserved gene families represents only a very rough sketch of the common inheritance of all modern life.

Summary

For most of human history, the living world around us was classified by what we could see. Genome sequencing has radically changed our view of life on the planet, and we now realize that living things fall into three broad domains: bacteria, archaea, and eukaryotes. The organisms in the first two domains are largely invisible to our naked eye, and many of them cannot yet be grown in a laboratory—being known only by their DNA sequences. But they make up the vast majority of life's evolutionary diversity, including species that can obtain all their energy and nutrients from inorganic chemical sources—such as the reactive mixtures of minerals released at hydrothermal vents on the ocean floor—the sort of diet that may

TABLE 1–1 The Number of Gene Families, Classified by Function, Common to All Three Domains of the Living World

Information processing		Metabolism	
Translation	63	Energy production and conversion	19
Transcription	7	Carbohydrate transport and metabolism	16
DNA replication, recombination, and repair	13	Amino acid transport and metabolism	43
Cellular processes and signaling		Nucleotide transport and metabolism	15
Cell-cycle control, mitosis, and meiosis	2	Coenzyme transport and metabolism	22
Defense mechanisms	3	Lipid transport and metabolism	9
Signal-transduction mechanisms	1	Inorganic ion transport and metabolism	8
Cell wall/membrane biogenesis	2	Secondary metabolite biosynthesis, transport, and catabolism	5
Intracellular trafficking and secretion	4	Poorly characterized	
Post-translational modification, protein turnover, chaperones	8	General biochemical function predicted; specific biological role unknown	24

For the purpose of this analysis, gene families are defined as “universal” if they are represented in the genomes of at least two diverse archaea (*Archaeoglobus fulgidus* and *Aeropyrum pernix*), two evolutionarily distant bacteria (*Escherichia coli* and *Bacillus subtilis*), and one eukaryote (yeast, *Saccharomyces cerevisiae*). (Data from R.L. Tatusov et al., *Science* 278:631–637, 1997; R.L. Tatusov et al., *BMC Bioinformatics* 4:41, 2003; and the COGs database at the US National Library of Medicine.)

have nourished the first living cells more than 3.5 billion years ago. The eukaryotes (whose cells are larger and contain a variety of membrane-bound organelles) evolved later in evolutionary history and are consequently less diverse as a group than either the bacteria or archaea. Eukaryotes, which include all plants and animals, are the organisms most familiar to us, and they are the main focus of this textbook.

Many of the genes within a single organism or species show strong family resemblances in their DNA sequences, implying that they originated from the same ancestral gene through gene duplication and divergence. Family resemblances (homologies) are also clear when gene sequences are compared between different species, and more than 200 gene families have been so highly conserved that they can be recognized as common to most species from all three domains of the living world, suggesting they were present in the ancestral cell from which all life evolved. Given the DNA sequence of a newly discovered gene in any organism, it is therefore often possible to deduce the gene's function from the known function of a homologous gene in a better-studied organism.

EUKARYOTES AND THE ORIGIN OF THE EUKARYOTIC CELL

Eukaryotic cells, in general, are bigger and more elaborate than bacterial and archaeal cells, and their genomes are bigger and more elaborate, too. The greater cell size is accompanied by radical differences in cell structure and function: in particular, eukaryotes contain a diverse set of intracellular **organelles**—discrete membrane-enclosed subcompartments and large membraneless macromolecular assemblies—each with a distinct composition and function. Some eukaryotic cells live independent lives as single-cell organisms. Others live in multicellular assemblies—indeed, all of the more complex multicellular organisms on Earth, including plants, animals, and fungi, are formed from eukaryotic cells.

We begin by discussing how eukaryotic cells are organized and how they might have evolved from more ancient prokaryotes. We then briefly consider how eukaryotic genomes differ from those of prokaryotes, as well as how the cells in

multicellular organisms become differently specialized as an embryo develops, so as to contribute to the welfare of the organism as a whole.

Eukaryotic Cells Contain a Variety of Organelles

By definition, eukaryotic cells keep almost all their DNA in a membrane-enclosed internal compartment—the *nucleus*, which is usually the most conspicuous organelle (**Figure 1-21**). The long DNA polymers in the nucleus are packaged with proteins to form *chromosomes*, which only become visible in a light microscope when they condense in preparation for cell division. The *nuclear envelope*, a double layer of membrane, surrounds the nucleus and separates the nuclear DNA from the *cytoplasm*, which, in a eukaryotic cell, includes everything between the plasma membrane and the nucleus. As shown in the figure, the nuclear envelope is perforated by *nuclear pores*, which are channels formed by protein complexes that mediate the two-way traffic of large molecules between the nucleus and the cytoplasm.

Eukaryotic cells have many other features that set them apart from bacterial and archaeal cells. They are typically 10–30 times bigger in linear dimension and 1000–10,000 times larger in volume than a typical prokaryotic cell. They have an elaborate *cytoskeleton* in the cytoplasm, consisting of several types of protein filaments (see Figure 1-21) that, together with the many proteins that attach to them, form a network of girders, ropes, and motors that gives the cell mechanical strength and performs various other functions: when the cell divides, for example, the cytoskeleton reorganizes and pulls the replicated chromosomes apart and distributes them equally to the two daughter cells. In the case of animal cells and some free-living, single-cell eukaryotes, the cytoskeleton controls cell shape

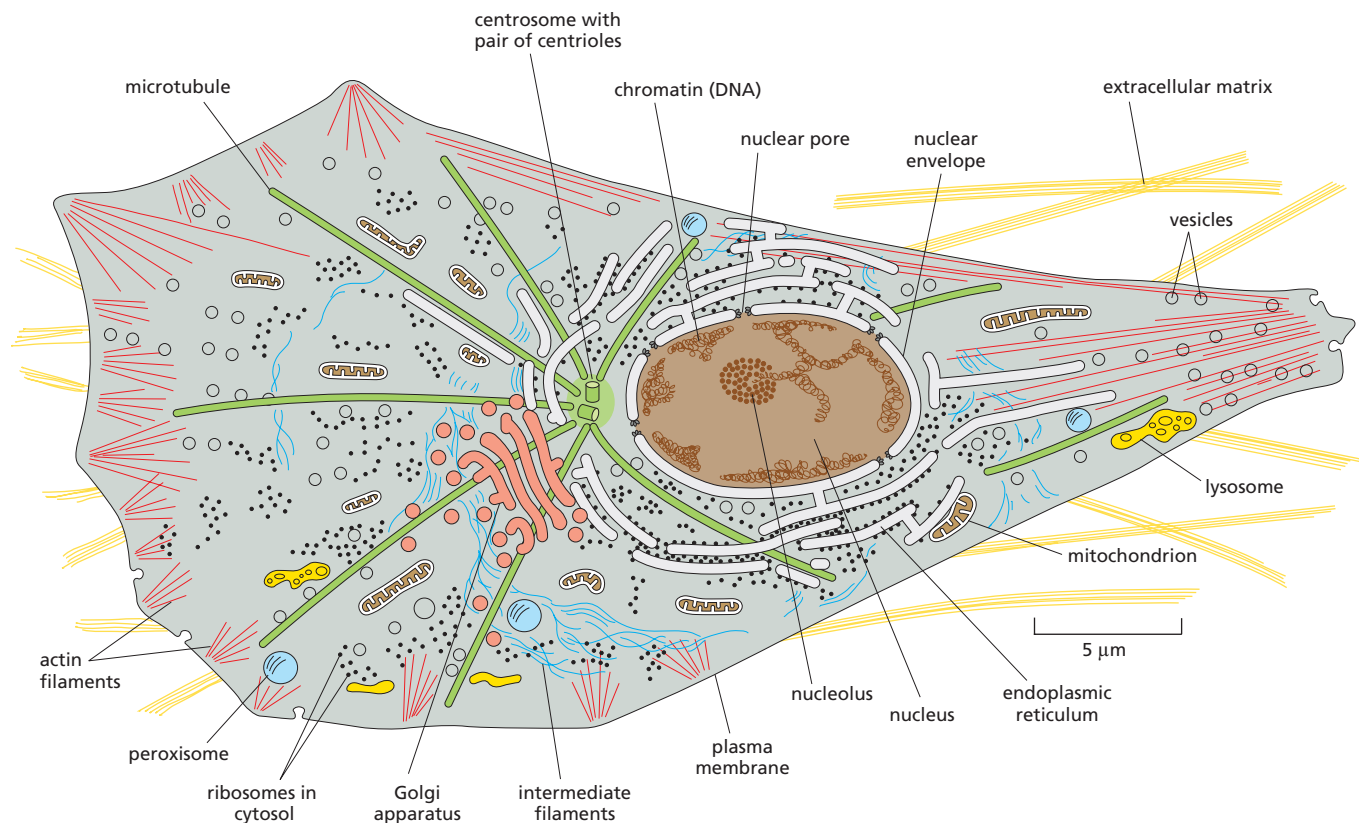


Figure 1-21 The major features of eukaryotic cells. The drawing depicts the major contents of a typical animal cell seen in cross section, but almost all the same components are found in plant cells and fungi, as well as in single-cell eukaryotes. The cytoskeleton (discussed in Chapter 16) consists of three types of protein filaments: actin filaments (*red*), microtubules (*green*), and intermediate filaments (*blue*). Plant cells (not shown) contain chloroplasts in addition to the components shown here; they also have a rigid external *cell wall* that contains cellulose surrounding their plasma membrane, which means they are largely immobile. The interior of cells is, in reality, much more crowded than depicted in this simplified diagram.

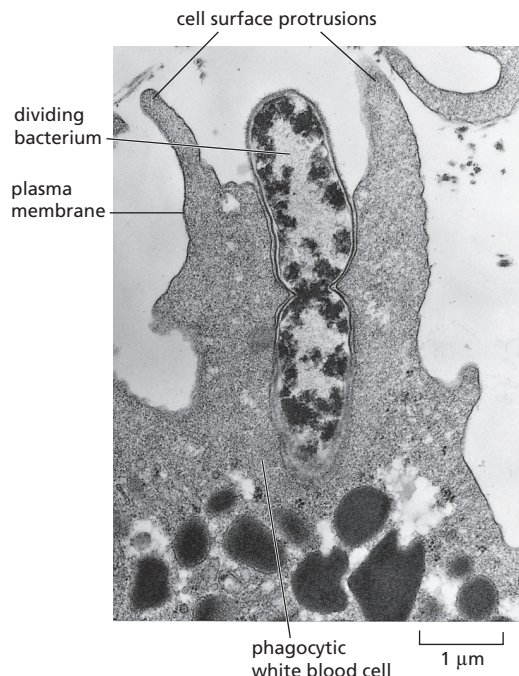


Figure 1–22 Phagocytosis. An electron micrograph of a mammalian phagocytic white blood cell (a neutrophil) ingesting a bacterium that is in the process of dividing. Only the part of the cell that is extending surface protrusions to engulf the bacterium is shown. (Courtesy of Dorothy Bainton.)

and drives and guides cell movements (**Movie 1.1**). Lacking the kind of tough cell wall characteristic of bacteria and archaea, these eukaryotic cells can change their shape rapidly, in some cases enabling them to move and engulf other cells and small objects by a process called *phagocytosis* (**Figure 1–22**).

There are many other membrane-enclosed organelles in eukaryotic cells. Unlike the nucleus, most of them are enclosed by single membranes. The most extensive organelle is the *endoplasmic reticulum* (ER), which is where most cell membrane components are made, along with materials destined for secretion to the outside of the cell. The *Golgi apparatus* receives these molecules from the ER and modifies and packages them for secretion or transport to another cell compartment. *Lysosomes* are small irregularly shaped organelles in which intracellular digestion occurs. *Peroxisomes* are small vesicles where hydrogen peroxide is used to inactivate toxic molecules.

A continual exchange of materials occurs between these single-membrane-enclosed organelles, mediated mainly by small *transport vesicles* that pinch off from the membrane of one organelle and fuse with that of another. To connect the eukaryotic cell with its surroundings, a similar vesicle-mediated exchange goes on continually at the cell surface. Here, portions of the plasma membrane pinch in to form intracellular vesicles that carry material captured from the external medium into the cell—a process called *endocytosis*; and in the reverse process, called *exocytosis*, vesicles from inside the cell fuse with the plasma membrane and release their contents to the exterior (**Figure 1–23**).

Besides the nucleus, there are two other eukaryotic cell organelles that are enclosed in double membranes—*mitochondria* and, in plant cells and algae, *chloroplasts*. Mitochondria take up oxygen and harness energy from the oxidation of food molecules, such as sugars and fats, to produce most of the ATP (adenosine triphosphate) that powers the cell's activities. *Chloroplasts* perform photosynthesis in plant cells and algae, using the energy of sunlight to synthesize carbohydrates from atmospheric CO₂ and water, delivering these energy-rich products to the host cell as food. In many eukaryotic cells, roughly half of the cytoplasm is occupied by membrane-enclosed organelles. The surrounding fluid is called the *cytosol*. It contains *ribosomes*, which translate RNAs into proteins, and it is also where most of the cell's other metabolic reactions take place.

In addition to the membrane-enclosed organelles just described, eukaryotic cells contain a variety of smaller organelles that lack membranes. Instead,

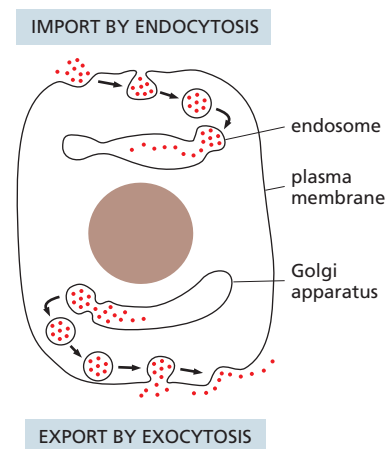


Figure 1–23 Endocytosis and exocytosis across the plasma membrane. Eukaryotic cells import extracellular materials by endocytosis and secrete intracellular materials by exocytosis. The endocytosed material is first delivered to single-membrane-enclosed organelles called *endosomes*, discussed in Chapter 12.

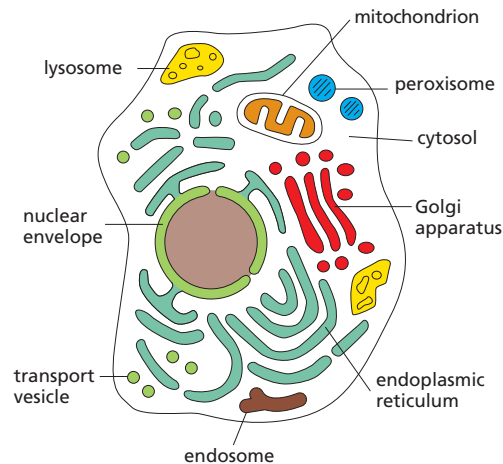


Figure 1–24 Membrane-enclosed organelles are distributed throughout the eukaryotic cell cytoplasm. The membrane-enclosed organelles, shown in different colors, are each specialized to perform a different function. The cytoplasm that fills the space outside of these organelles is called the cytosol.

multiple components that work together are held in close proximity in **bio-molecular condensates**, which will be described in Chapters 3, 6, and 12. An example is the nucleolus, where ribosome assembly takes place (see Figure 1–21). **Figure 1–24** summarizes the membrane-enclosed organelles in eukaryotic cells, each of which will be discussed in detail in later chapters.

Mitochondria Evolved from a Symbiotic Bacterium Captured by an Ancient Archaeon

A fundamental question in both evolution and cell biology is how did the first eukaryotic cell arise? The evidence suggests that it happened when an archaeal and a bacterial cell merged about 2 billion years ago, in a world that had contained only prokaryotes for more than 1.5 billion years.

All eukaryotic cells contain (or at one time did contain) mitochondria (**Figure 1–25**). Mitochondria are similar in size to small bacteria, and both reproduce by dividing. The mitochondria contain their own DNA, with genes that resemble bacterial genes; they also contain their own ribosomes and translation factors that resemble those in bacteria. These and other similarities between mitochondria and present-day bacteria provide strong evidence that mitochondria evolved from an aerobic bacterium (one that harvested energy by combining electrons derived from foodstuffs with oxygen gas) that was captured

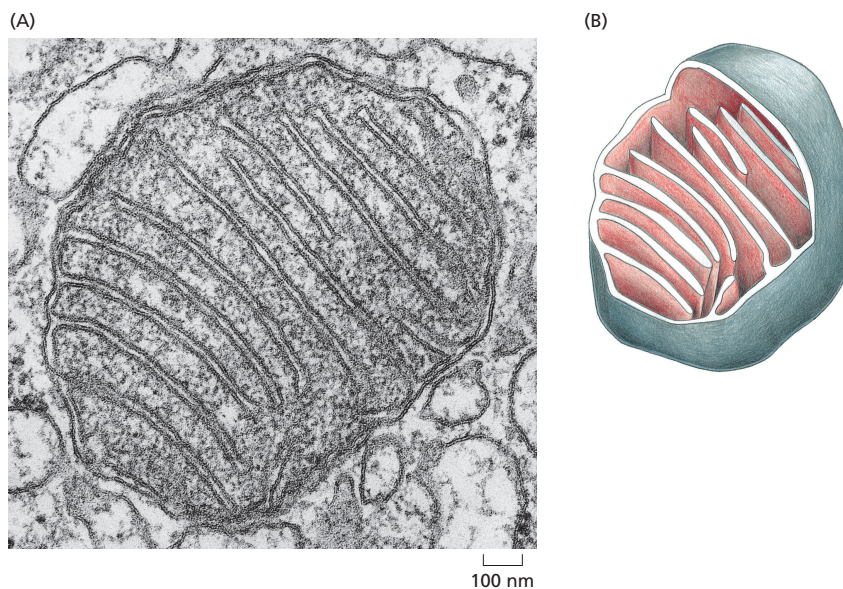
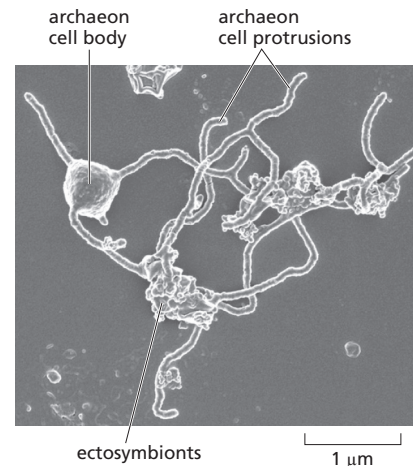


Figure 1–25 A mitochondrion. (A) An electron micrograph of the organelle seen in cross section. (B) A drawing of a mitochondrion with part of it cut away to show the three-dimensional structure (**Movie 1.2**). Note the smooth outer membrane and the convoluted inner membrane, which houses the proteins that generate ATP from the oxidation of food molecules. (A, courtesy of Daniel S. Friend and by permission of E.L. Bearer.)

Figure 1–26 A scanning electron micrograph of an Asgard archaeon in culture. This anaerobic cell proliferates very slowly, doubling only about every 20 days (compared to every half hour or so for the bacterium *E. coli*). It can be seen to extend elaborate membranous protrusions from its surface—including “blebs” and unique branched and unbranched structures. These protrusions are intimately associated with two other species—one bacterial, one archaeal—that were isolated with the Asgard strain as ectosymbionts, as indicated. The scientists had maintained the deep marine sediment under anaerobic conditions in a bioreactor for more than 2000 days, mimicking conditions of the seabed, and they attempted to culture samples from this bioreactor under a range of different conditions. Only after many years and repeated subculturing were they able to isolate this archaeon with its ectosymbionts. (From H. Imachi et al., *Nature* 577:519–525, 2020.)



by an anaerobic cell. These two cells (and their descendants) were then able to evolve an endosymbiotic relationship, providing mutual metabolic support within a common cytoplasm.

There are also good reasons to believe that the ancestral capturing cell was an archaeon. As we have seen, the genomes of present-day archaea encode many proteins that are characteristic of present-day eukaryotic cells. Those archaea with the most eukaryotic-like genes belong to the Asgard lineage, first identified by sequencing DNA fragments obtained from the seabed. But no living example had been seen or cultivated—until very recently, when, after a heroic 12-year-long isolation procedure, the first Asgard archaeon was propagated in culture. This remarkable, living, anaerobic archaeon looked like no other prokaryote—having long branching protrusions—and it seemed to live in an ectosymbiotic relationship with another bacterium and another archaeon, both of which were isolated with it (Figure 1–26). The discovery of the strange Asgard archaeon provides a glimpse of how an ancient archaeon might eventually have captured an aerobic bacterium to initiate the eukaryotic lineage, with a hypothetical pathway being illustrated in Figure 1–27 and discussed in detail in Chapter 12.

Chloroplasts Evolved from a Symbiotic Photosynthetic Bacterium Engulfed by an Ancient Eukaryotic Cell

Chloroplasts (Figure 1–28) perform photosynthesis in plant cells and algae, using the energy of sunlight to synthesize their own “food” (in the form of carbohydrates) from atmospheric CO₂ and water. Like mitochondria, they are enclosed in double membranes, have their own “circular” genomes, and reproduce by dividing. They almost certainly evolved from a symbiotic photosynthetic bacterium that was captured by an ancient eukaryotic cell that already possessed mitochondria. This bacterium may have been captured by phagocytosis, a frequent process in eukaryotes (see Figure 1–22).

Whereas some single-cell eukaryotes are hunters that live by capturing other cells and eating them (see Figure 1–34), a eukaryotic cell equipped with chloroplasts has no need to chase after other cells as prey; it is nourished by the captive chloroplasts it has inherited from its ancient eukaryotic ancestors. Correspondingly, plant cells, although they possess the cytoskeletal equipment for movement, have lost the ability to change shape rapidly and to engulf other cells

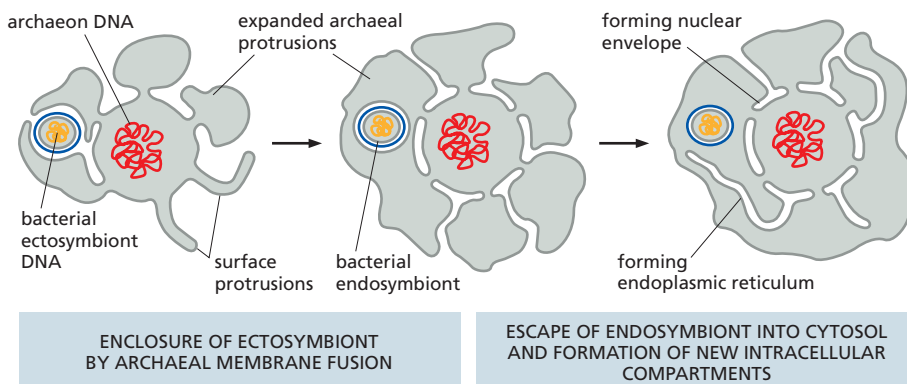


Figure 1–27 A possible model for some early steps in eukaryotic cell evolution.

In this model, the surface protrusions of an ancient Asgard archaeon expanded and surrounded an ectosymbiotic aerobic bacterium to create a symbiotic relationship between the two types of cells. Eventually, the protrusions fused with one another, trapping the bacterium as an *endosymbiont* in the archaeon cytoplasm, where it was initially enclosed by an internal membrane derived from the archaeon's plasma membrane (the bacterium itself retaining its own membranes). At some point, the endosymbiont escaped from the enclosing archaeon-derived membrane and entered the cytosol, where it eventually evolved into a mitochondrion—with both its DNA and membranes derived from the engulfed bacterium. As shown, it is postulated that the internal archaeon membranes generated by this mechanism of protrusion expansion and fusion progressively formed both the nucleus and single-membrane-enclosed organelles, such as the endoplasmic reticulum. The evidence for this general type of model of eukaryotic-cell evolution is discussed further in Chapter 12. (Adapted from H. Imachi et al., *Nature* 577:519–525, 2020, and from D.A. Baum and B. Baum, *BMC Biol.* 12:76–92, 2014.)

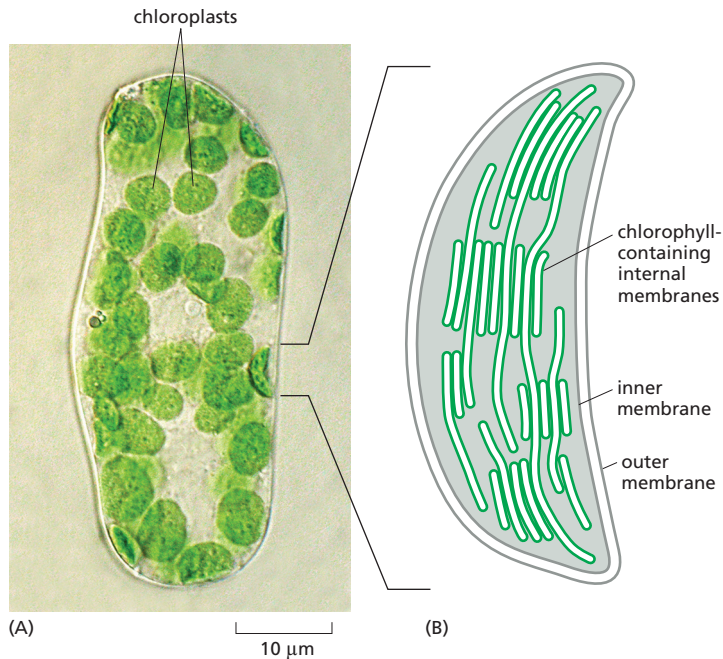


Figure 1–28 Chloroplasts. In plant cells and single-celled photosynthetic eukaryotes, these organelles capture the energy of sunlight. (A) A light micrograph of a single cell isolated from a leaf of a flowering plant, showing the green chloroplasts (**Movie 1.3** and see **Movie 14.10**). (B) A drawing of one chloroplast, showing the highly folded system of internal membranes containing the chlorophyll molecules that absorb light. (A, courtesy of Preeti Dahiya.)

by phagocytosis. Instead, they wrap themselves in a tough, protective cell wall. If some eukaryotic cells can be viewed as hunters, then one might view plant cells as having given up hunting for farming.

Fungi represent yet another eukaryotic way of life. Fungal cells, like animal cells, possess mitochondria but not chloroplasts; they have a tough outer wall that limits their ability to move rapidly or to take up other cells. Fungi, it seems, have turned from hunters into scavengers. Other cells secrete nutrient molecules or release them after death, and fungi feed on these leavings—often performing whatever digestion is necessary extracellularly, by secreting digestive enzymes to the exterior.

Eukaryotes Have Hybrid Genomes

As just discussed, the genetic information of eukaryotic cells has a hybrid origin—from an ancestral anaerobic archaeon and from the bacteria it adopted as endosymbionts (**Figure 1–29**). Most of this genetic information (DNA) is stored in the nucleus, although a small amount remains inside the organelles that evolved

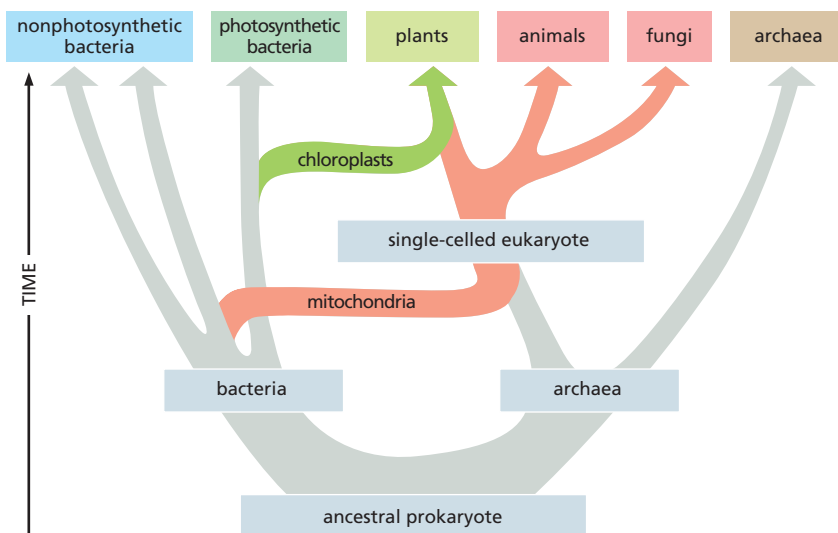


Figure 1–29 A model for the evolution of eukaryotic cells in the tree of life. All living cells are thought to have evolved from an ancestral prokaryotic cell (the last universal common ancestor) between 3.5 and 3.8 billion years ago). Many millions of years later, it seems an anaerobic archaeon acquired an aerobic bacterial symbiont, which evolved into mitochondria (see **Figure 1–27**). Later still, a mitochondria-containing eukaryotic cell acquired a photosynthetic bacterium, which evolved into chloroplasts. Mitochondria are essentially the same in plants, animals, and fungi, indicating that they were acquired before these three lineages diverged about 1.5 billion years ago.

from the captured bacteria—the mitochondria and, in plant and algal cells, the chloroplasts. When the mitochondrial DNA and the chloroplast DNA are separated from the nuclear DNA and individually sequenced, both the mitochondrial and chloroplast genomes are found to be cut-down versions of the corresponding bacterial genomes: in a human cell, for example, the mitochondrial genome consists of only 16,569 nucleotide pairs, and it codes for only 13 proteins plus a set of 24 RNAs involved in protein synthesis.

Many of the genes that are missing from the mitochondria and chloroplasts have not been lost; instead, they have moved from the endosymbiont genomes into the DNA of the host-cell nucleus. Thus the nuclear DNA of animals contains many genes coding for proteins that serve essential functions inside the mitochondria; in plants and algae, the nuclear DNA contains many genes specifying proteins required in chloroplasts. In both cases, the DNA sequences of these nuclear genes still show clear evidence of their bacterial origins.

Eukaryotic Genomes Are Big

Natural selection has evidently favored mitochondria with small genomes. By contrast, the nuclear genomes of most eukaryotes seem to have been free to enlarge. Perhaps the eukaryotic way of life has made large size an advantage: predatory cells, for example, typically need to be bigger than their prey, and cell size generally increases in proportion to genome size. Whatever the reason, the genomes of most eukaryotes have become hundreds of times larger than those of bacteria and archaea (Figure 1–30).

The freedom to be extravagant with DNA has had profound implications. Eukaryotes not only have more genes than prokaryotes; they also have vastly more DNA that does not code for protein or RNA. The human genome contains about 700 times as many nucleotide pairs as the genome of a typical bacterium such as *E. coli*, but it contains only about 4.5 times as many protein-coding genes because a much greater proportion of the human genome does not code for protein (~98.5% compared to 11% in *E. coli*). The estimated genome sizes and gene numbers for a few selected eukaryotes are compared with the bacterium *E. coli* in Table 1–2; we will discuss shortly how each of these organisms serves as a model organism.

Eukaryotic Genomes Are Rich in Regulatory DNA

As discussed in Chapter 4, much of our nonprotein-coding DNA is almost certainly dispensable “junk,” retained during evolution like a mass of old papers because, when there is little pressure to keep an archive small, it is easier to retain

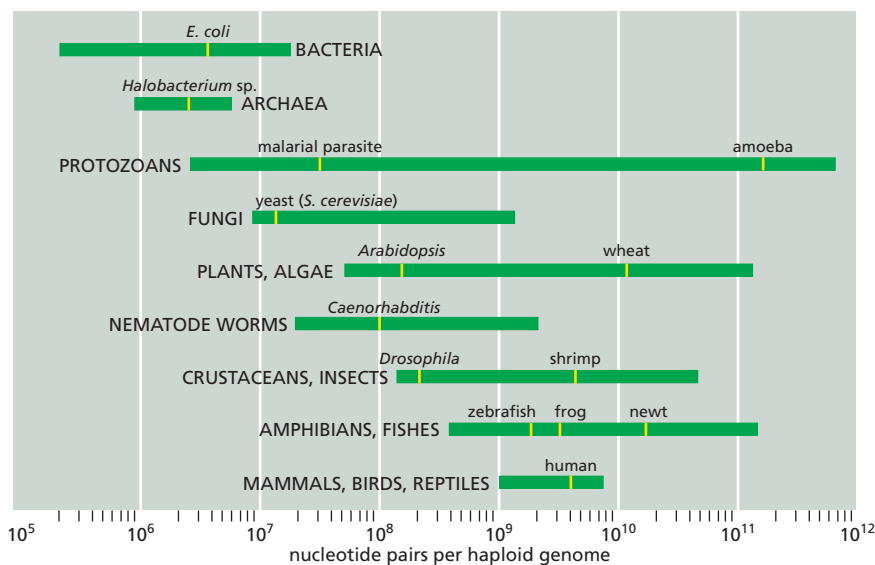


Figure 1–30 Genome sizes compared. Genome size is measured in nucleotide (base) pairs of DNA per haploid genome, that is, per single copy of the genome. (The body cells of sexually reproducing, multicellular organisms such as ourselves are generally diploid: they contain two copies of the genome, one inherited from the mother, the other from the father.) Note that closely related organisms can vary widely in the quantity of DNA in their genomes (as indicated by the length of the green bars), even though they contain similar numbers of protein-coding genes. (Data from T.R. Gregory, 2021. Animal Genome Size Database: www.genomesize.com.)

TABLE 1–2 Some Model Organisms and Their Genomes

Organism	Approximate genome size* (nucleotide pairs)	Approximate number of protein-coding genes**
<i>Escherichia coli</i> (bacterium)	4.6×10^6	4300
<i>Saccharomyces cerevisiae</i> (yeast)	12.5×10^6	6600
<i>Caenorhabditis elegans</i> (roundworm)	100×10^6	20,000
<i>Arabidopsis thaliana</i> (plant)	135×10^6	27,000
<i>Drosophila melanogaster</i> (fruit fly)	180×10^6	14,000
<i>Danio rerio</i> (zebrafish)	1400×10^6	26,000
<i>Mus musculus</i> (mouse)	2800×10^6	20,000
<i>Homo sapiens</i> (human)	3100×10^6	20,000

*Genome size includes an estimate for the amount of highly repeated, noncoding DNA sequence, which does not appear in genome databases.

**There are also genes that code for functional RNA molecules that do not code for proteins.

everything than to sort out the valuable information and discard the rest. Certain exceptional eukaryotic species, such as the puffer fish, bear witness to the profligacy of their relatives; they have somehow managed to rid themselves of large quantities of nonprotein-coding DNA, and, yet, they appear similar in structure, behavior, and fitness to related species that have vastly more such DNA.

Even in compact eukaryotic genomes such as that of the puffer fish, there is more nonprotein-coding DNA than protein-coding DNA. As in all eukaryotic organisms, at least some of the noncoding DNA certainly has important functions. In particular, it regulates the expression of genes. With this *regulatory DNA*, eukaryotes have evolved distinctive, highly sophisticated ways of controlling when and where a gene is brought into play. Elaborate mechanisms for gene regulation are especially crucial for the formation and function of complex multicellular organisms, which have many different cell types, each with different functions, as we now discuss.

Eukaryotic Genomes Define the Program of Multicellular Development

The cells in an individual animal or plant are extraordinarily varied. Blood cells, skin cells, bone cells, nerve cells—they seem as dissimilar as any cells could be (Figure 1–31). Yet all these cell types are the descendants of a single fertilized egg cell, and all (with very minor exceptions) contain identical copies of the genome of the species.

The differences result from the way in which the cells make selective use of their genetic instructions according to their developmental history and the cues they receive from their surroundings in a developing embryo. The DNA is not just a shopping list specifying the molecules that every cell must have, and the cell is not an assembly of all the items on the list. Rather, the cell behaves as a multipurpose machine, with sensors to receive environmental signals and with highly developed abilities to call different sets of genes into action according to signals it receives. The genome in each cell is big enough to accommodate the information that specifies an entire multicellular organism, but in any individual cell only part of that information is used.

Many genes in the eukaryotic genome code for proteins that regulate the activities of other genes, a topic discussed in detail in Chapter 7. Most of these regulatory genes encode *transcription regulators* that act by binding, directly or



Figure 1–31 Cell types can vary enormously in size and shape. A human nerve cell is compared here with a human neutrophil, a type of white blood cell. Both are drawn to scale.



Figure 1-32 Genetic control of the program of multicellular development. The role of a regulatory gene is demonstrated in the snapdragon *Antirrhinum*. In this example, a mutation in a single gene coding for a regulatory protein causes leafy shoots to develop in place of flowers: because the regulator protein has been changed, the cells adopt characters that would be appropriate to a different location in the normal plant. The mutant is on the left, the normal plant on the right. (Courtesy of Enrico Coen and Rosemary Carpenter.)

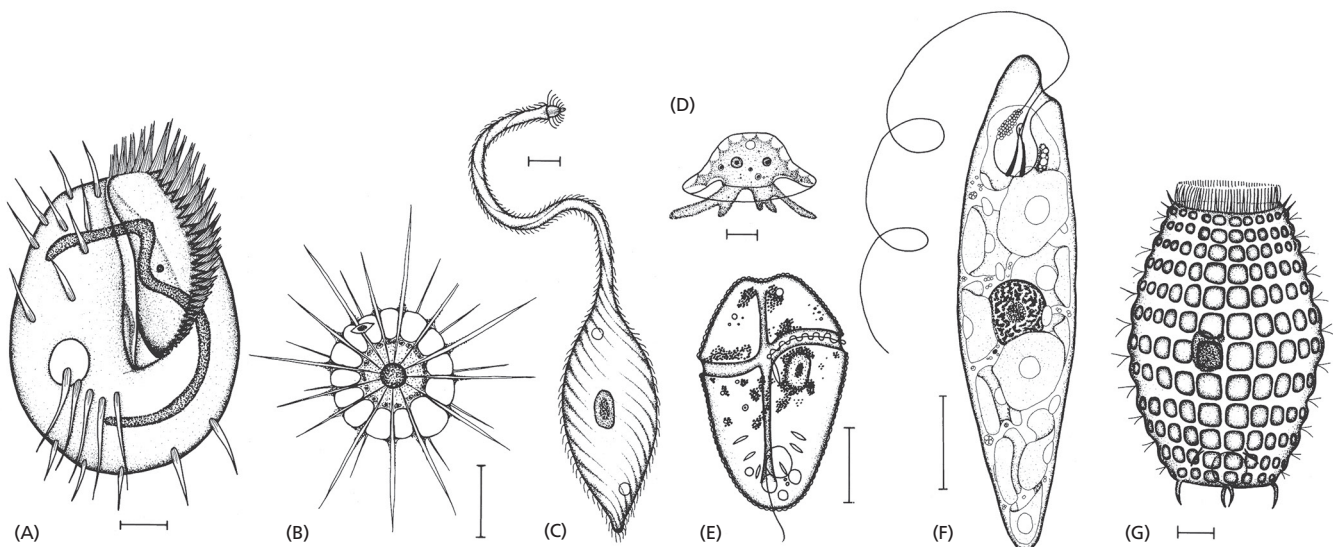
indirectly, to the many different DNA sequences that control which genes are to be expressed, and at what levels. Eukaryotic genomes also produce many noncoding RNA molecules; as the name implies, they are not translated into protein but control gene expression in a variety of ways. The expanded genome of eukaryotes therefore not only specifies the hardware of the cell but also stores the software that controls how that hardware is used (**Figure 1-32**).

Cells do not just passively receive signals; rather, they actively exchange signals with their neighbors. Thus, in a developing multicellular organism, an internal control system governs each cell that has different consequences depending on the messages exchanged. The outcome, astonishingly, is a precisely patterned array of cells of different types, each displaying a character appropriate to its position in the multicellular structure.

Many Eukaryotes Live as Solitary Cells

Many species of eukaryotic cells lead a solitary life. As we have seen, some of these single-cell organisms are hunters, some are photosynthesizers, still others are scavengers. **Figure 1-33** conveys something of the astonishing variety of single-cell eukaryotes, whose anatomy can be remarkably elaborate, including

Figure 1-33 An assortment of single-cell eukaryotes. The drawings are done to different scales, but in each case the scale bar represents 10 μm . The organisms in (A), (C), and (G) are ciliates; (B) is a heliozoan; (D) is an amoeba; (E) is a dinoflagellate; and (F) is a euglenoid. (Courtesy of Michael Sleight, from M.A. Sleight, *Biology of Protozoa*. Edinburgh: Edward Arnold, 1973.)



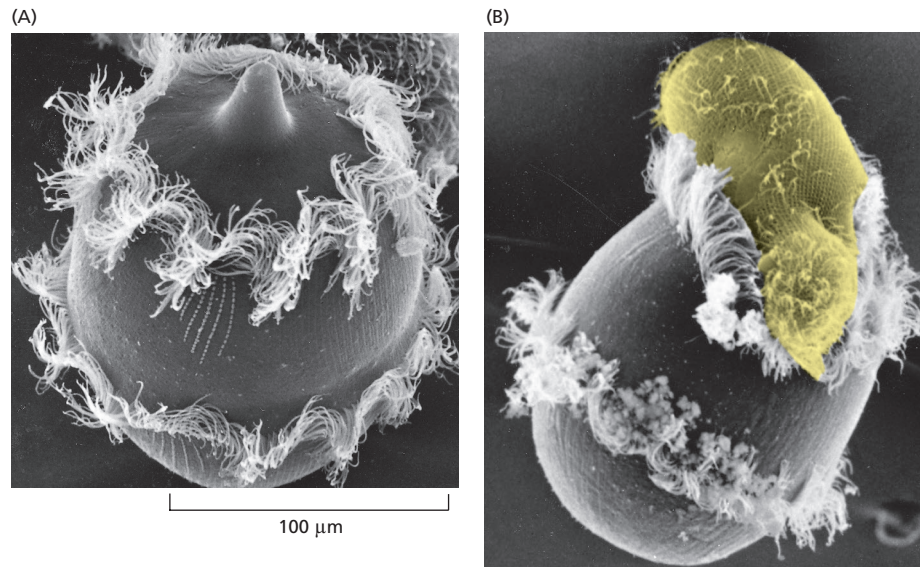


Figure 1-34 A single-cell eukaryote that eats other cells. (A) *Didinium* is a carnivorous protist, belonging to the group of *ciliates*. (A protist is defined as a free-living, single-cell, mobile eukaryote.) *Didinium* has a globular body, about 150 μm in diameter, encircled by two fringes of cilia—sinuous, whiplike appendages that beat continually; its front end is flattened except for a single protrusion, rather like a snout. (B) A *Didinium* engulfing its prey. It normally swims around in the water at high speed by means of the synchronous beating of its cilia. When it encounters a suitable prey, usually another type of protist, it releases numerous small paralyzing darts from its snout region. Then, it attaches to and devours its prey (artificially colored *yellow*) by phagocytosis, inverting like a hollow ball to engulf its victim, which can be almost as large as itself. (Courtesy of D. Barlow.)

such structures as sensory bristles, photoreceptors, sinuously beating cilia, leglike appendages, mouth parts, stinging darts, and muscle-like contractile bundles. Although they are single cells, they can be as intricate, as versatile, and as complex in their behavior as many multicellular organisms (**Figure 1-34**, **Movie 1.4**, and **Movie 1.5**).

Humans tend to focus on plants and animals, while neglecting single-cell eukaryotes because (along with bacteria and archaea) they are microscopic. But thanks to DNA comparisons, we now know that the genomes of single-cell eukaryotes are far more evolutionarily diverse than those of multicellular animals and plants, meaning that animals and plants arose relatively late in the complex and fascinating eukaryotic pedigree. With genome data, we can position the many different single-celled eukaryotes in the tree of life and identify our closest relatives (**Figure 1-35**). Scientists are using this information to probe the origins of multicellularity, with a focus on what these strange creatures can tell us about our own evolutionary past.

Summary

Eukaryotic cells, by definition, keep most of their DNA in a separate membrane-enclosed compartment—the nucleus. They have, in addition, an elaborate set of other organelles, each carrying out different functions, such as the oxidation of food-derived molecules and production of ATP in mitochondria. Eukaryotic cells also contain a cytoskeleton for structural support and movements. There is compelling evidence that mitochondria and, in plants and algae, chloroplasts evolved from captured symbiotic bacteria, which explains why these organelles contain their own DNA and ribosomes.

Eukaryotic cells typically have 3–8 times as many protein-coding genes as bacteria and archaea and often a thousand times more noncoding DNA. Although much of this DNA is probably unimportant, some of it allows for great complexity in the regulation of gene expression, as required for the construction of complex multicellular organisms containing many different cell types.

MODEL ORGANISMS

Because all cells appear to have descended from a common ancestor, whose fundamental properties have been conserved through evolution, the knowledge gained from the study of one organism contributes to our understanding of all others, including ourselves. It turns out that certain organisms are much more accessible than others for study in the laboratory. Some reproduce rapidly and

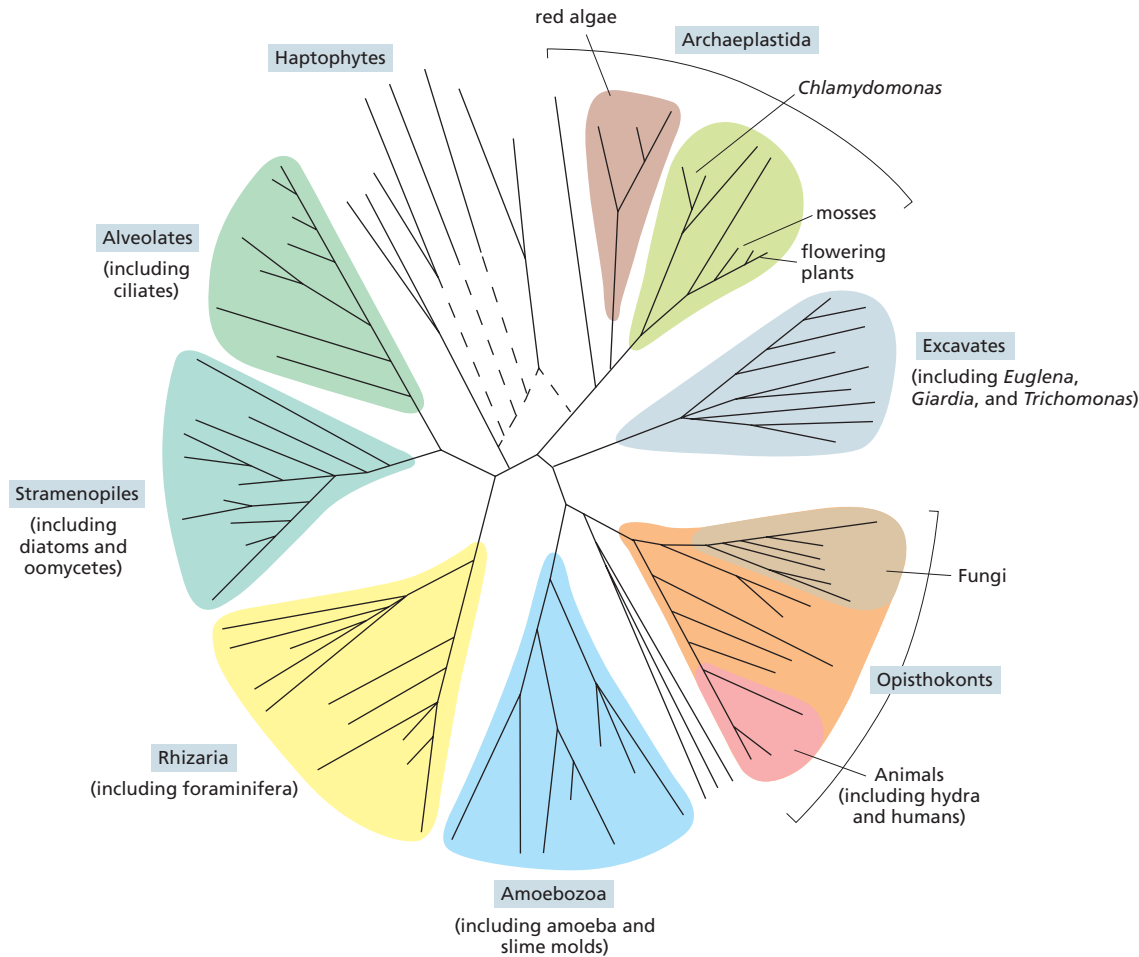


Figure 1–35 A eukaryotic tree of life based on genome comparisons. The lengths of the lines are proportional to the extents of genome diversity, with broken lines indicating uncertain relationships. Note that animals and plants are separated by many single-cell species (including the Excavates), suggesting that multicellularity arose independently several times during eukaryotic evolution. (Adapted from F. Burki, *Cold Spring Harb. Perspect. Biol.* 6:a016147, 2014.)

are easily manipulated using powerful genetic techniques. Others are transparent and readily develop in the laboratory from a fertilized egg to a multicellular organism, so that one can readily trace how their cells behave to produce internal tissues and organs.

Over time, different groups of biologists have focused on studying a few chosen species, which allows their knowledge and research tools to be pooled to gain a deeper understanding than could be achieved if their efforts were spread over many different organisms. Although the list of these representative, model organisms is continually expanding, a few stand out in terms of the breadth and depth of information that has been accumulated about them over the years—knowledge that has been essential for our understanding of how all cells work. In this section, we examine some of these organisms and review the benefits that each offers to the study of cell biology and, in many cases, to the promotion of human health. We begin with a discussion of some especially powerful strategies that scientists have developed to understand the cell, and we shall see how these approaches dictated the choice of model organisms.

Mutations Reveal the Functions of Genes

Without additional information, no amount of gazing at genome sequences will reveal the functions of genes. We may recognize that gene B is like gene A, but how do we discover the function of gene A in the first place? And even if we know

the function of gene A, how do we test whether the function of gene B is truly the same as the sequence similarity suggests? How do we connect the world of abstract genetic information that was introduced in the previous sections with the world of living cells and organisms?

The analysis of gene functions depends on two highly complementary approaches: biochemistry and genetics. Biochemistry directly examines the functions of purified molecules, such as the protein and RNA produced from a specific gene: first we obtain that molecule from an organism and then study its chemical activities in detail. In contrast, genetics starts with the study of mutants: we either find or make an organism in which the specific gene is altered, and we then examine the effects on the mutant organism's structure and performance (**Figure 1-36**). When combined with biochemistry, careful studies of an organism (and its isolated cells) mutated for a particular protein or RNA molecule can reveal the biological role of that molecule.

Biochemistry and genetics, used in combination with cell biology, provide a powerful way to connect genes and molecules directly to cell and organism structure and function. In recent years, DNA sequence information and the powerful tools of molecular biology have greatly accelerated progress in this endeavor. From sequence comparisons, we can often identify particular subregions within a gene that have been conserved nearly unchanged over the course of evolution. These subregions are often the most important parts of the gene in terms of function. We can test their individual contributions to the gene's function by creating in the laboratory mutations of specific sites within the subregion or by constructing artificial hybrid genes that combine part of one gene with parts of another. Organisms can be engineered to make either the RNA or protein specified by the gene in large quantities to facilitate biochemical analysis. Specialists in molecular structure can determine the three-dimensional conformation of the gene product, revealing the exact position of every atom in it. Biochemists can determine how each of the parts of the genetically specified molecule contributes to its chemical behavior and function in a test tube. Cell biologists determine the many other molecules that interact with the molecule of interest and where all these molecules are located within a cell. And they also analyze the behavior of cells that are engineered to express a mutant version of the gene.

There is, however, no one simple universal recipe for discovering a gene's function. We may discover, for example, that the product of a given gene is an enzyme that catalyzes a certain chemical reaction, and yet have no idea how or why that reaction is important to the organism. The functional characterization of each new gene product or family of gene products, unlike the description of the gene sequences, presents a fresh challenge to the biologist's ingenuity. Moreover, we will never fully understand the function of a gene until we learn its role in the life of the organism, which means studying whole organisms, not just isolated molecules or cells.

Molecular Biology Began with a Spotlight on One Bacterium and Its Viruses

Because living organisms are so complex, the more we learn about any particular species, the more attractive it becomes as an object for further study. Each discovery in such a chosen organism raises new questions and provides new tools with which to tackle general biological questions. For this reason, large communities of biologists have become dedicated to studying different aspects of the same **model organism**.

In the early days of molecular biology, the chosen model was the bacterium *Escherichia coli* (*E. coli*—see Figure 1-11B). This small, rod-shaped cell normally lives in the gut of humans and other vertebrates, but it can be grown easily in a simple nutrient broth in a culture bottle or dish, where under favorable conditions, it can reproduce every 20 minutes or so. It adapts to variable chemical conditions and can evolve by mutation and selection at a remarkable speed.

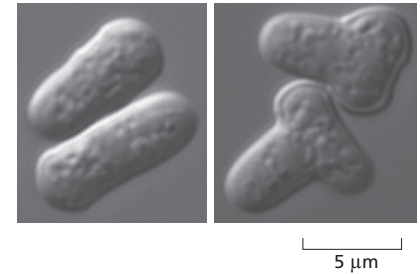


Figure 1-36 An alteration in organism shape resulting from a gene mutation. Scanning electron micrographs (discussed in Chapter 9) of a normal yeast (of the species *Schizosaccharomyces pombe*) compared with a mutant yeast, where a change in a single gene has converted the cell from a cigar shape (*left*) to a T shape (*right*). The mutant gene therefore has a function controlling cell shape. But how, in molecular terms, does the gene product perform that function? That is a harder question, and it needs biochemical analysis to answer it. (Courtesy of Kenneth Sawin and Paul Nurse.)

Also of special early interest were a few of the viruses that infect this bacterium—inasmuch as their much smaller genomes made them even easier to analyze in detail.

Viruses are small packets of genetic material that have evolved as parasites that depend on the reproductive and biosynthetic machinery of the host cells they infect. Viruses are not strictly alive, because they depend on the machinery of their host cells for their reproduction. Although we now know that viruses are the most abundant—in terms of sheer numbers—of all the biological entities on this planet, they are too small to be seen in the light microscope. For this reason, they were completely missed until the end of the nineteenth century, when a few viruses were identified as infectious agents that pass through filters that trap bacteria, but are retained by the even-finer filters that allow large molecules to pass. Only with the invention of the electron microscope could viruses finally be visualized as tiny particles with defined shapes and sizes. We now know that viruses consist of many families, with different families having distinct structures and modes of replication (discussed in Chapters 6 and 23).

Those viruses that infect bacteria are called *bacteriophages*, and two that infect *E. coli* have played critical roles as model organisms that advanced our understanding of molecular cell biology. Detailed genetic analyses of these two viruses, bacteriophage lambda and bacteriophage T4, came first, followed by biochemistry that used the analysis of mutant genes to identify and characterize specific proteins of interest. Geneticists, for example, generated and then characterized more than a hundred different mutant genes in bacteriophage T4, a large virus with a double-strand DNA genome (Figure 1–37). Sets of T4 genes that encode components of the head and the tail of the bacteriophage were identified, allowing biochemical studies to reveal important principles of biological assembly processes. Similarly, a set of T4 genes that geneticists showed were essential for T4

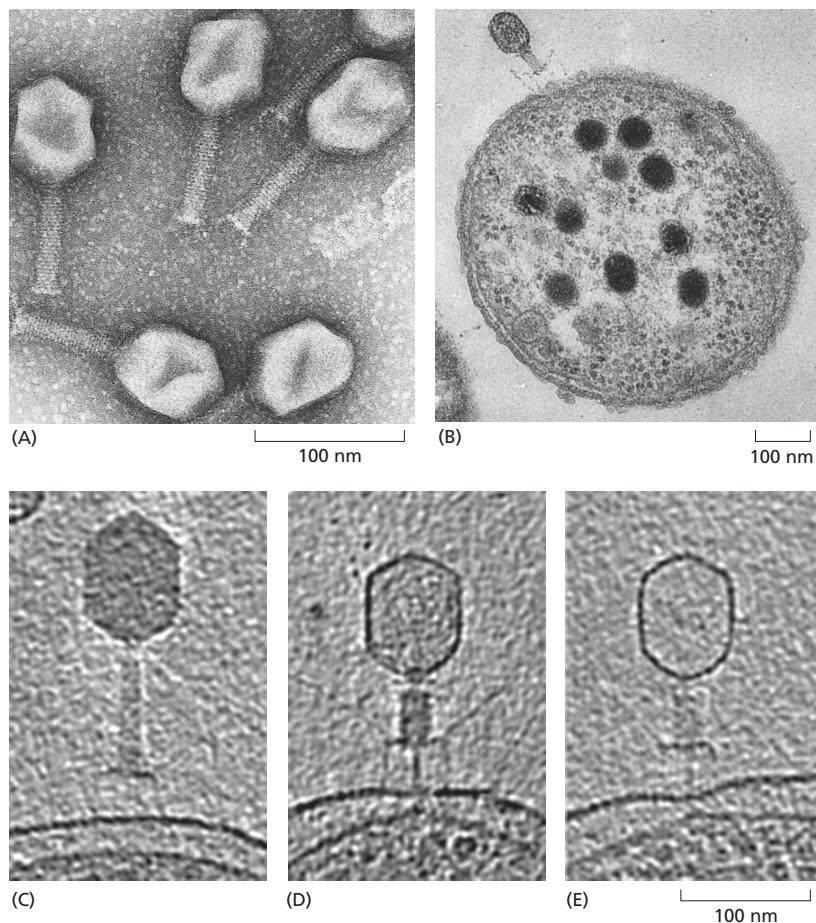


Figure 1–37 The T4 bacteriophage.

(A) An electron micrograph of particles of the T4 bacteriophage, a virus that infects *E. coli* bacteria. The hexagonal head of the virus contains the viral DNA; the tail contains the apparatus for injecting the DNA into a host bacterium. (B) A cross section of an *E. coli* bacterium with a T4 bacteriophage attached to its surface. The large dark objects inside the bacterium are the assembling heads of new T4 particles. When the particles are mature, the bacterium will burst open and release them. (C–E) The process of DNA injection into the bacterium, as visualized in unstained, frozen samples by cryo-electron microscopy (discussed in Chapter 9). (C) Attachment begins. (D) Attached state during DNA injection. (E) Virus head has emptied its entire DNA into the bacterium. (A, courtesy of James Paulson; B, courtesy of Jonathan King and Erika Hartwig from G. Karp, *Cell and Molecular Biology*, 2nd ed. New York: John Wiley & Sons, 1999. With permission from John Wiley & Sons; C–E, courtesy of Ian Molinieux, University of Texas at Austin, and Jun Liu.)

DNA replication allowed those proteins to be purified, so that biochemists could decipher the central mechanisms of DNA replication in a test tube. In the same way, it was extensive studies of bacteriophage lambda that led to our early understanding of transcription regulators and gene regulatory networks (see Panel 7-1 on pp. 404–405 and Figure 7-43).

We now know that these two bacteriophages have many close relatives distributed throughout the biosphere. Relatives of bacteriophage T4, for example, are abundant in the ocean, where they infect the ubiquitous marine cyanobacteria. As a whole, ocean viruses are present in enormous numbers, estimated at 10^{30} . If lined up end to end, they would extend beyond our nearest galaxies; they kill approximately 20% of the total ocean microbial biomass per day. Because these viruses have such a huge role in nutrient recycling, they profoundly affect Earth's ecology.

Although not themselves living cells, viruses often serve as vectors for gene transfer between cells. A virus will replicate in one cell, emerge from it with a protective wrapping, and then enter and infect another cell, which may be of the same or different species. Often, the infected host cell is killed by the massive proliferation of virus particles inside it, but sometimes the viral DNA, instead of directly generating new virus particles, may persist in its host for many cell generations as a relatively innocuous passenger—either as a separate intracellular fragment of DNA, known as a *plasmid*, or as a DNA sequence inserted into the cell's own genome. In their travels, viruses can accidentally pick up fragments of DNA from the genome of one host cell and ferry them into another cell. Such transfers of genetic material are very common in prokaryotes.

Many bacterial and archaeal species have a remarkable capacity to take up even nonviral DNA molecules from their surroundings and thereby capture the genetic information these molecules carry. By this route or by virus-mediated gene transfer, bacteria and archaea in the wild can acquire genes from neighboring cells relatively easily. Genes that confer resistance to an antibiotic or an ability to produce a toxin, for example, can be transferred from species to species and provide the recipient bacterium with a selective advantage, greatly enhancing its rate of spread. In this way, new and sometimes dangerous strains of antibiotic-resistant bacteria have been observed to evolve in the bacterial ecosystems that inhabit hospitals or various niches in the human body. On a longer time scale, the results can be even more profound; it has been estimated that at least 18% of all the genes in the present-day genome of *E. coli* have been acquired by horizontal transfer from another species within the past 100 million years.

The Focus on *E. coli* as a Model Organism Has Accelerated Many Subsequent Discoveries

The standard laboratory strain *E. coli* K-12 has a genome of approximately 4.6 million nucleotide pairs contained in a single circular molecule of DNA that codes for about 4300 different kinds of proteins (Figure 1-38). In molecular terms, we probably have a more complete understanding of *E. coli* than of any other living organism. Most of our understanding of the fundamental mechanisms of life—for example, how cells replicate their DNA or how they decode the instructions represented in the DNA to direct the synthesis of specific RNAs and proteins—initially came from studies of *E. coli* and its viruses. This is because the basic genetic mechanisms have turned out to be highly conserved throughout evolution and are essentially the same in our own cells as in *E. coli*.

It should be noted that, as with other bacteria, different strains of *E. coli*, though classified as members of a single species, differ genetically to a much greater degree than do different varieties of an organism such as a plant or animal. One *E. coli* strain may possess many hundreds of genes that are absent from another, and the two strains could have as little as 50% of their genes in common. These differences are largely the result of rampant horizontal gene transfer, characteristic of this and many other bacterial and archaeal species.

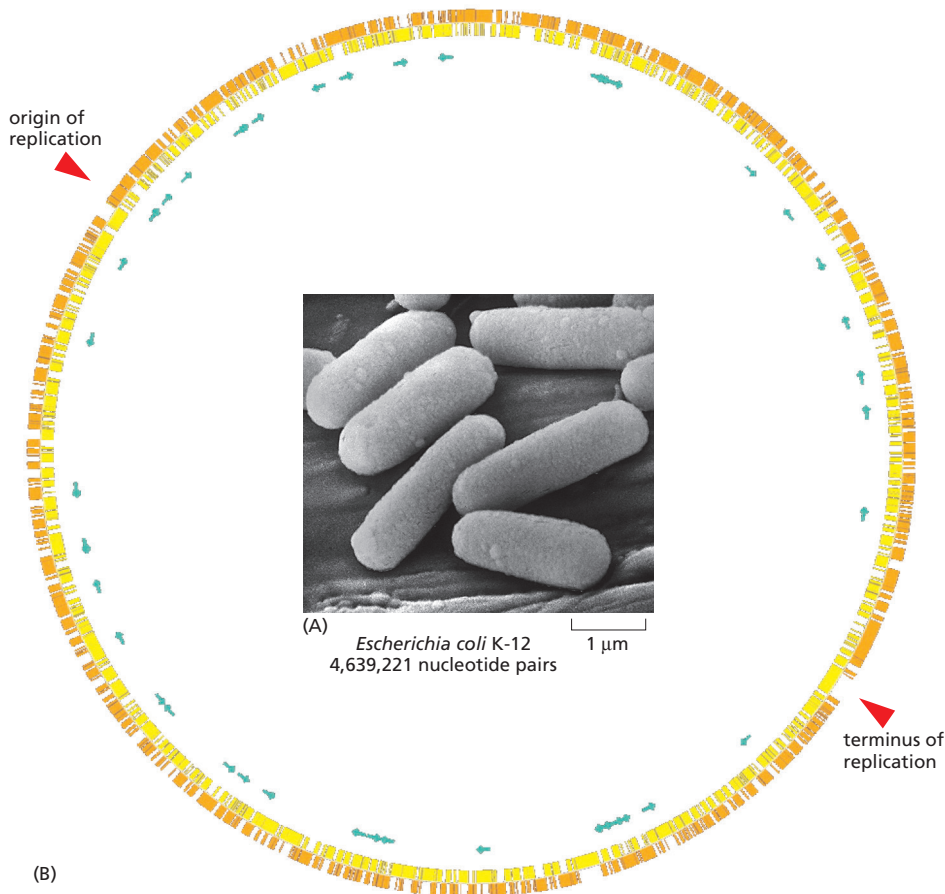


Figure 1–38 The genome of *E. coli*. (A) A cluster of *E. coli* cells viewed in a scanning electron microscope. (B) A diagram of the genome of *E. coli* strain K-12. The diagram is circular because the DNA of *E. coli*, like that of most other bacteria, forms a single, closed loop. Protein-coding genes are shown as yellow or orange bars, depending on the DNA strand from which they are transcribed; RNA molecules produced from nonprotein-coding genes are indicated by green arrows, with the arrowheads indicating their direction of transcription. Some genes are transcribed from one strand of the DNA double helix (in a clockwise direction in this diagram), others from the other strand (counterclockwise). The origin and terminus of DNA replication are marked with red arrowheads. (A, Dr. Tony Brain & David Parker/Science Source; B, adapted from F.R. Blattner et al., *Science* 277:1453–1462, 1997.)

A Yeast Serves as a Minimal Model Eukaryote

The molecular and genetic complexity of eukaryotes is daunting, and biologists need to concentrate their limited resources on a small number of selected model organisms to unravel this complexity.

To analyze the internal workings of the eukaryotic cell without the additional problems of multicellular development, it makes sense to use a single-cell species that is as simple as possible. The popular choice for this role of minimal model eukaryote has been the yeast *Saccharomyces cerevisiae* (Figure 1-39)—the same species that is used by brewers of beer and bakers of bread.

S. cerevisiae is a small, single-cell member of the kingdom of fungi and, in terms of its genome sequence, much more closely related to animals than to plants (see Figure 1-35). It is robust and easy to grow in a simple nutrient medium. Like other fungi, it has a tough cell wall, is relatively immobile, and possesses mitochondria but not chloroplasts. When nutrients are plentiful, it grows and divides about every hundred minutes. It can reproduce either vegetatively (that is, by ordinary cell division, or mitosis), or sexually: two yeast cells that are *haploid* (possessing a single copy of each chromosome, $n = 1$) can fuse to create a cell that is *diploid* (containing two copies of each chromosome, $n = 2$), and the diploid cell can undergo *meiosis* (a reduction division) to produce cells that are once again

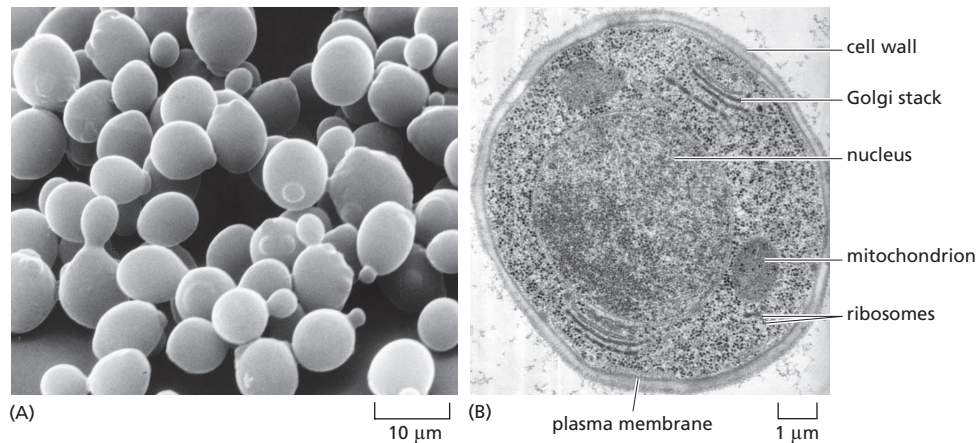


Figure 1-39 The yeast *Saccharomyces cerevisiae*. (A) A scanning electron micrograph of a cluster of yeast cells. This species is also known as budding yeast, because it proliferates by forming a protrusion, or bud, that enlarges and then separates from the mother cell. Many cells with buds are visible in this micrograph. (B) An electron micrograph of a cross section of a yeast cell, showing its plasma membrane and thick cell wall, as well as some of its intracellular organelles. (A, courtesy of Ira Herskowitz and Eric Schabtach; B, courtesy of Andrew Staehelin.)

haploid (**Figure 1-40**). In contrast to most animals, this yeast can therefore proliferate either sexually or asexually, a choice that an experimenter can make simply by changing the growth conditions.

In addition to these features, the yeast has a further property that makes it a convenient organism for genetic studies: its genome, by eukaryotic standards, is exceptionally small (see Table 1-2), yet it suffices for all the basic tasks that every eukaryotic cell must perform. Mutants are available for every gene, and thus the consequence of missing each gene—one by one—can be observed under any environmental condition using the high-throughput procedures described in Chapter 8. Over the past 50 years, extensive studies of yeast cells carried out by many laboratories have provided keys to crucial “eukaryotic-only” processes. These include the cell-division cycle (the critical chain of events by which the nucleus and all the other components of a cell are duplicated and parceled out to create two daughter cells from one) and meiosis (the process through which an organism’s reproductive cells are formed). In addition, important insights into eukaryotic chromosome structure, the organization of the nucleus, the mechanisms of gene expression, the formation of organelles, and the ways that proteins are secreted from cells have come out of the work on yeasts. Many of these fundamental processes are so similar between yeasts and humans that a human homolog of a yeast protein will often faithfully carry out its functions when artificially expressed in yeast cells.

The Expression Levels of All the Genes of an Organism Can Be Determined

The complete genome sequence of *S. cerevisiae* consists of approximately 12,500,000 nucleotide pairs, including the small contribution (about 78,500 nucleotide pairs) of the mitochondrial DNA. This total is only about 2.7 times as much DNA as there is in *E. coli*, and it codes for only about 1.5 times as many distinct proteins (see Table 1-2). The way of life of *S. cerevisiae* is similar in many ways to that of a bacterium, and it seems that this yeast has likewise been subject to selection pressures (for rapid proliferation, for example) that have kept its genome compact.

Knowledge of the complete genome sequence of any organism—be it a yeast or a human—opens up new perspectives on the workings of the cell: many things that once seemed impossibly complex now seem to be within our grasp. Using techniques described in Chapter 8, it is possible, for example, to monitor simultaneously, the amount of mRNA produced from every gene in the yeast genome under any environmental condition. It is also possible to determine in real time

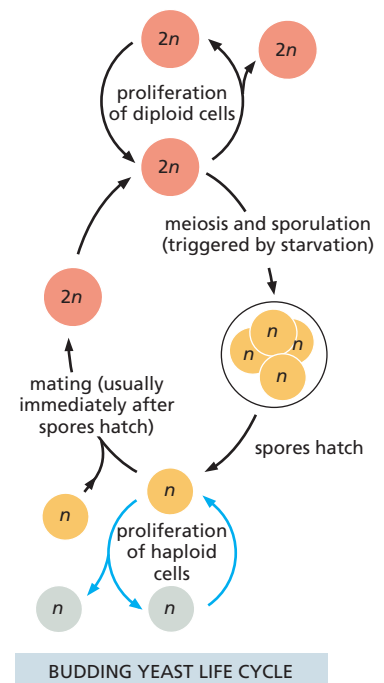


Figure 1-40 The reproductive cycles of the yeast *S. cerevisiae*. Depending on environmental conditions and on details of the genotype, cells of this species can exist in either a diploid ($2n$) state, with a double chromosome set, or a haploid (n) state, with a single chromosome set. The diploid form can either proliferate by ordinary cell-division cycles (mitosis) or undergo meiosis to produce haploid cells. The haploid form can either proliferate by ordinary cell-division cycles or undergo sexual fusion with another haploid cell to become diploid. Meiosis is triggered by starvation, and it gives rise to spores—haploid cells in a dormant state, resistant to harsh environmental conditions.

how the pattern of gene activity changes when conditions change. This type of analysis can be repeated with mRNA prepared from mutant cells lacking any gene we care to test, and, in this way, the influence of that gene on the expression of all other genes can be observed. Although pioneered in yeast, this approach now provides a way to reveal the entire system of controls that govern gene expression in any organism, as long as its genome sequence is known and it can be manipulated genetically.

Arabidopsis Has Been Chosen as a Model Plant

The large multicellular organisms that we see around us—the plants and animals—seem fantastically varied, but, as we have seen, they are much closer to one another in their evolutionary origins, and more similar in their basic cell biology, than the great host of microscopic single-celled organisms we have been discussing. Thus, while bacteria and archaea are separated by perhaps 3.5 billion years of evolution, vertebrates and insects are separated by about 700 million years, fish and mammals by about 450 million years, and the different species of flowering plants by only about 150 million years (see Figure 1–35).

Because of the close evolutionary relationship between all flowering plants (see Figure 1–35), we can, once again, gain insight into the cell and molecular biology of this whole class of organisms by focusing on just one or a few species for detailed analysis. Out of the nearly 400,000 known species of flowering plants, molecular biologists have chosen to concentrate their efforts on a small weed in the cabbage family, the common wall cress *Arabidopsis thaliana* (Figure 1–41), which can be grown indoors in large numbers and produces thousands of offspring per plant after 8–10 weeks. *Arabidopsis* has a total genome size of approximately 135 million nucleotide pairs, about 10 times the size of the yeast genome (see Table 1–2).

Work on *Arabidopsis* has provided a deep understanding of numerous key features of plants, including the mechanisms that cause flower development and its coordination with the seasons, the ability to grow toward sunlight, cell-to-cell signaling by hormones, and the special type of innate immune system that plants use to ward off pathogens. Comparison of the developmental programs between plants and animals has also highlighted some common principles, thereby allowing a glimpse into the basic logic through which large, highly differentiated, multicellular organisms evolved from single-cell ancestors.

The World of Animal Cells Is Mainly Represented by a Worm, a Fly, a Fish, a Mouse, and a Human

Although plants make up 80% of the biomass on Earth and animals make up less than 0.4% (see Figure 1–14), animals account for the majority of all named species of living organisms, and they are by far the most intensely studied. Five species have emerged as the foremost model organisms for molecular, cell, and developmental biological studies. In order of increasing body size, they are the nematode worm *Caenorhabditis elegans*, the fly *Drosophila melanogaster*, the zebrafish *Danio rerio*, the mouse *Mus musculus*, and the human, *Homo sapiens*. Genome sequences from many different individuals within each species have been determined.

Caenorhabditis elegans (Figure 1–42) is a small, harmless relative of the eelworm that attacks crops. With a life cycle of only a few days, an ability to survive in a freezer indefinitely in a state of suspended animation, a simple body

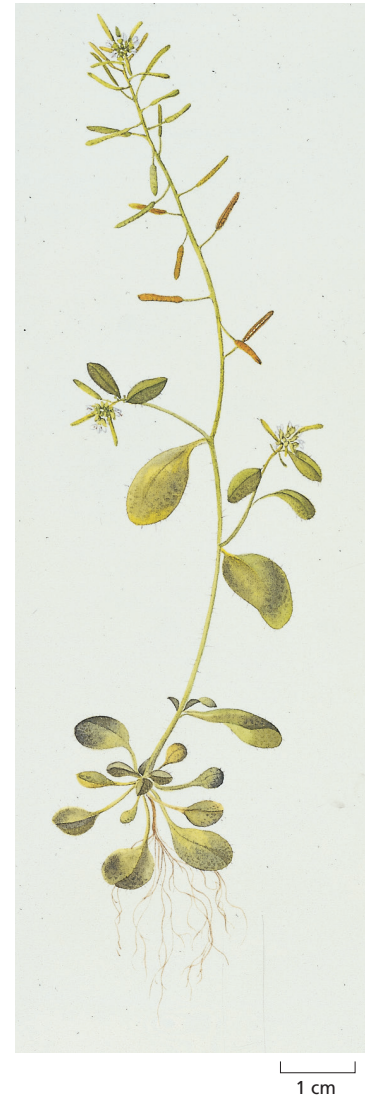


Figure 1–41 *Arabidopsis thaliana*, the plant chosen as the primary model for studying plant molecular genetics. (Courtesy of Toni Hayden, FLS; and the John Innes Foundation.)



Figure 1–42 *Caenorhabditis elegans*, the first multicellular organism to have its complete genome sequence determined. This nematode is only about 1 mm long and normally lives in the soil. Most individuals are hermaphrodites, producing both eggs and sperm. (Courtesy of Maria Gallegos, University of Wisconsin, Madison.)

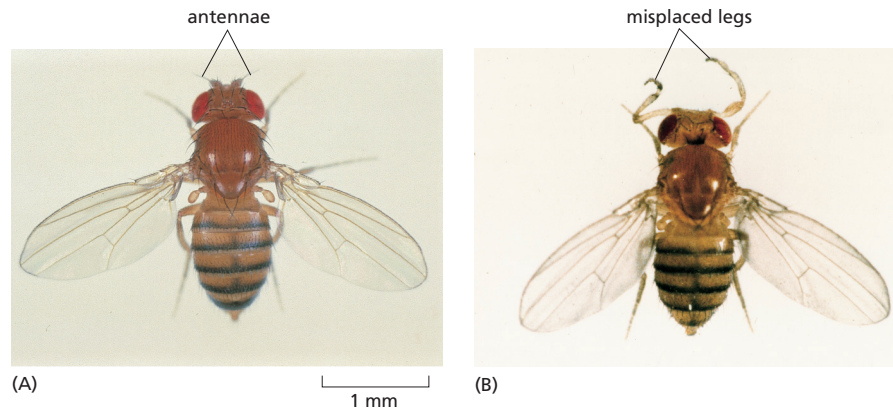


Figure 1-43 *Drosophila melanogaster*. (A) A normal adult *Drosophila* fruit fly. Molecular genetic studies of this fly have provided the main key to understanding how all animals develop from a fertilized egg into an adult. (B) A mutant adult fly, in which a mutation in a regulatory DNA sequence has caused genes for leg formation to be abnormally activated in positions normally reserved for antennae; as a result, legs have developed where antennae should be. (A, Edward B. Lewis, Courtesy of the Archives, California Institute of Technology; B, courtesy of Matthew Scott.)

plan, and an unusual life cycle that is well suited for genetic studies, it is an attractive model animal. *C. elegans* develops with clockwork precision from a fertilized egg cell into an adult worm with exactly 959 body cells (plus a variable number of egg and sperm cells)—an unusual degree of regularity for animal development. We now have a minutely detailed description of the sequence of events by which this development occurs, as the cells divide, move, and change their character according to strict and predictable rules (see Figure 21-42). The genome of about 100 million nucleotide pairs codes for about 20,000 proteins, and many mutants and other tools are available for testing gene functions. Although the worm has a body plan very different from our own, the conservation of biological mechanisms has been sufficient for the worm to be a model for many of the developmental and cell-biological processes that occur in the human body. Thus, for example, studies of the worm have been critical for understanding the molecular mechanisms that mediate and regulate the many cell deaths that help control animal-cell numbers, both in normal development and during human cancer growth. This crucial process, called *programmed cell death* or *apoptosis*, is the subject of Chapter 18. In addition, studies in *C. elegans* first revealed many fascinating features of RNA interference (discussed in Chapters 7 and 8). They have also provided key insights into the ways neurons make their proper connections (discussed in Chapter 21) and informed many additional areas of cell biology.

Studies in the Fruit Fly *Drosophila* Provide a Key to Vertebrate Development

The fruit fly *Drosophila melanogaster* (Figure 1-43) has been used as a model for animal genetic studies for longer than any other organism; in fact, the foundations of classical genetics were built to a large extent on studies of this insect. Nearly 100 years ago, for example, the fly provided definitive proof that genes—the abstract units of hereditary information at the time—are carried on chromosomes, whose behavior had been closely followed with the light microscope during eukaryotic cell division but whose function was at first unknown. The proof depended on one of the many features that make *Drosophila* especially convenient for molecular genetic studies—the giant chromosomes, which have a characteristic banded appearance that is visible in some of its cells (Figure 1-44). Specific changes in the hereditary information, manifest in families of mutant flies, were found to correlate exactly with the loss or alteration of specific bands in the giant chromosomes.

In more recent times, *Drosophila*, more than any other organism, has shown us how to trace the chain of cause and effect from the genetic instructions encoded in the chromosomal DNA to the structure of the adult multicellular body. *Drosophila* mutants with body parts strangely misplaced (Figure 1-43) or mispatterned provided the key to the identification and characterization of the genes required to make a properly structured body, with gut, limbs, eyes, and all the other parts in their correct places. Once these *Drosophila* genes were identified, scientists could identify homologous genes in vertebrates, and then test their functions



Figure 1-44 Giant chromosomes from salivary gland cells of *Drosophila*. Because many rounds of DNA replication have occurred without an intervening cell division, each of the chromosomes in these unusual cells contains more than 1000 identical double-strand DNA molecules, all aligned in register. This makes them easy to see in the light microscope, where they display a characteristic and reproducible pattern of bands. Specific bands can be identified as the locations of specific genes: a mutant fly with a region of the banding pattern missing or altered shows a phenotype reflecting loss of the genes in that region (not shown). Genes that are being transcribed at a high rate correspond to bands with a “puffed” appearance (black arrow). The bands stained dark brown in the micrograph are sites where a particular regulatory protein is bound to the DNA; the regulatory protein is identified by the binding of a specific antibody. (From R. Paro, *Trends Genet.* 6:416-421, 1990. With permission from Elsevier.)

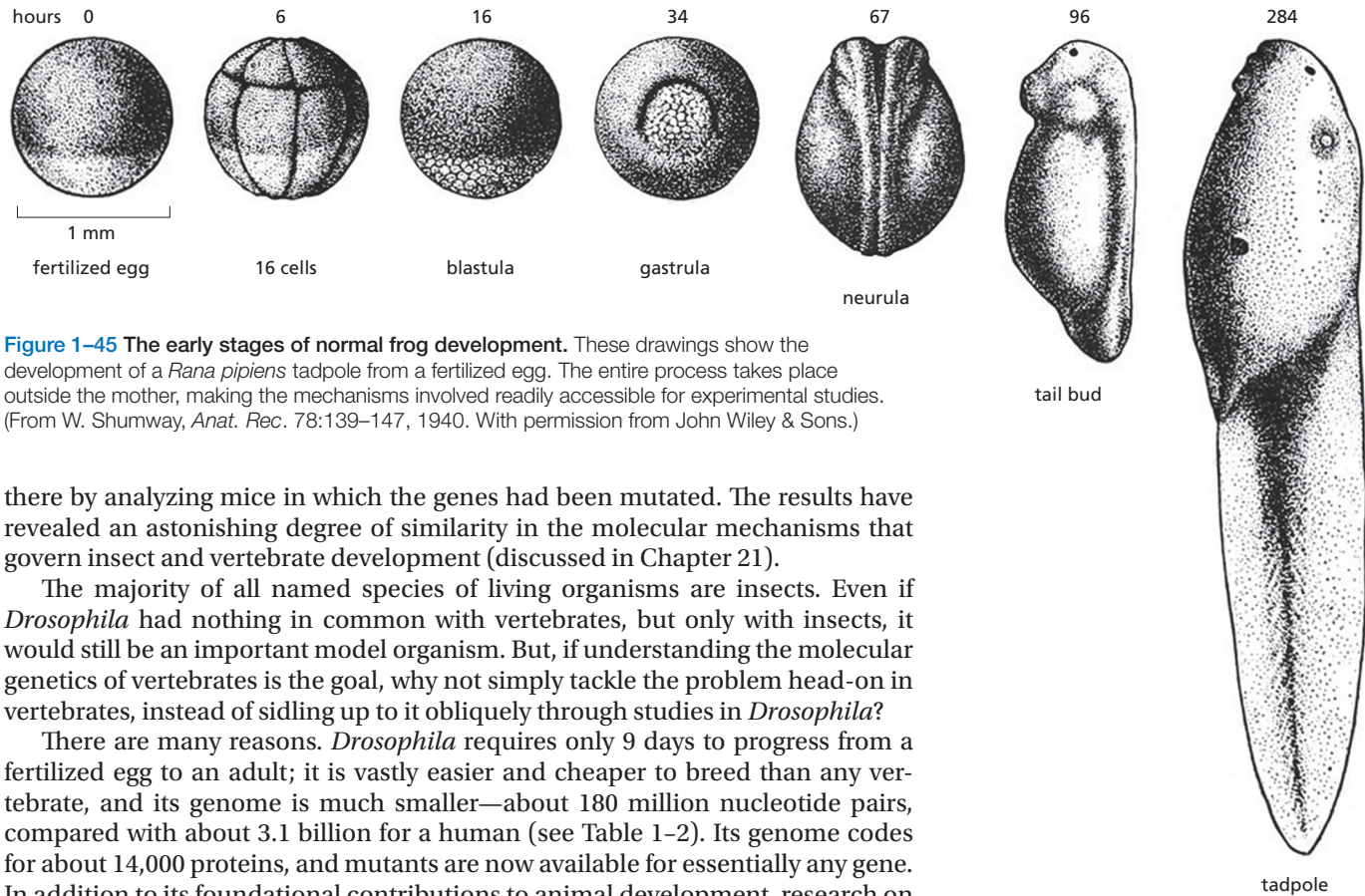


Figure 1-45 The early stages of normal frog development. These drawings show the development of a *Rana pipiens* tadpole from a fertilized egg. The entire process takes place outside the mother, making the mechanisms involved readily accessible for experimental studies. (From W. Shumway, *Anat. Rec.* 78:139–147, 1940. With permission from John Wiley & Sons.)

there by analyzing mice in which the genes had been mutated. The results have revealed an astonishing degree of similarity in the molecular mechanisms that govern insect and vertebrate development (discussed in Chapter 21).

The majority of all named species of living organisms are insects. Even if *Drosophila* had nothing in common with vertebrates, but only with insects, it would still be an important model organism. But, if understanding the molecular genetics of vertebrates is the goal, why not simply tackle the problem head-on in vertebrates, instead of sidling up to it obliquely through studies in *Drosophila*?

There are many reasons. *Drosophila* requires only 9 days to progress from a fertilized egg to an adult; it is vastly easier and cheaper to breed than any vertebrate, and its genome is much smaller—about 180 million nucleotide pairs, compared with about 3.1 billion for a human (see Table 1-2). Its genome codes for about 14,000 proteins, and mutants are now available for essentially any gene. In addition to its foundational contributions to animal development, research on *Drosophila* continues to uncover many other insights into biology, ranging from deeply conserved mechanisms that neutralize pathogens to ways that external stimuli from the environment are processed in the brain.

The Frog and the Zebrafish Provide Highly Accessible Vertebrate Models

Frogs have long been used to study the early steps of embryonic development in vertebrates. Because their eggs are big, easy to manipulate, and fertilized outside of the animal, the subsequent development of the early embryo can be easily followed (Figure 1-45). *Xenopus laevis*, the African clawed frog, continues to be an important model organism (Movie 1.6 and see Movie 21.1). Although the species is poorly suited for genetic analysis, cytoplasm isolated from unfertilized *Xenopus* eggs has the remarkable ability to recapitulate the formation of cellular structures and organelles in a test tube. These *egg extracts* allow powerful biochemical approaches to study such fundamental processes as the cell division cycle, described in Chapter 17.

The zebrafish *Danio rerio*, in contrast, is well suited for genetic analysis. Its genome is compact—only half as big as that of a mouse or a human (see Table 1-2)—and it has a generation time of only about 3 months, which is much

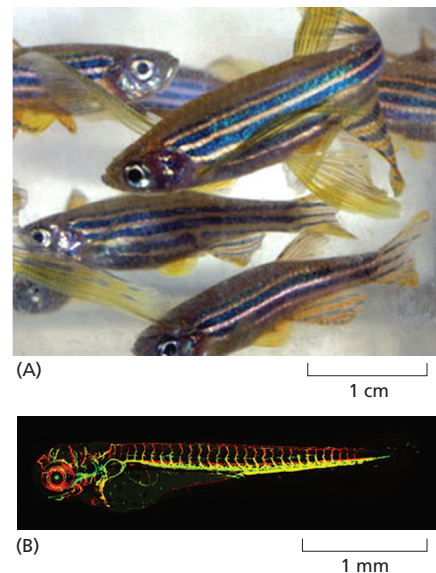


Figure 1-46 Zebrafish as a model for studies of vertebrate development. (A) These small, hardy tropical fish are found in many home aquaria and are convenient for laboratory genetic studies. They are ideal for developmental studies as their embryos develop outside of the mother and are transparent, so one can observe cells and internal structures in the living organism throughout its development from an egg to an adult. (B) In this fluorescence image of a 2-day-old embryo, a red fluorescent protein marks the developing blood vessels, and a green fluorescent protein marks the developing lymphatic vessels; regions where the two markers coincide appear yellow. (A, courtesy of Steve Baskauf; B, from H.M. Jung et al., *Development* 144:2070–2081, 2017. With permission from The Company of Biologists.)

shorter than that of *Xenopus laevis*. Many mutants are available, and genetic manipulation is relatively simple. The zebrafish has the added virtue that it is transparent for the first 2 weeks of its life, so behavior of specific tissues and individual cells can easily be followed in the living organism as it develops (Figure 1–46; see Movie 21.2). All this has made it an increasingly important model vertebrate, one that has been especially crucial for understanding the development of the heart and the circulatory system, as discussed in Chapter 22.

The Mouse Is the Predominant Mammalian Model Organism

In terms of genome size and function, cell biology, and molecular mechanisms, mammals are a highly uniform group of organisms. Even anatomically, the differences among mammals are chiefly a matter of size and proportions; it is hard to think of a human body part that does not have a counterpart in elephants and mice, and vice versa. Evolution plays freely with quantitative features, but it does not readily change the logic of the basic structure.

Mammals have typically about 1.5 times as many protein-coding genes as *Drosophila*, a genome that is about 16 times larger, and an adult body made up of millions or billions of times as many cells. For an exact measure of how closely mammalian species resemble one another genetically, we can compare the nucleotide sequences of corresponding (orthologous) genes or the amino acid sequences of the proteins that these genes encode. The results for individual genes and proteins vary widely. But typically, if we line up the amino acid sequence of a human protein with that of the orthologous protein from, say, an elephant, more than 80% of the amino acids are identical. A similar comparison between human and bird shows an amino acid identity of about 70%—because the bird and mammalian lineages have had longer to diverge than those of the elephant and the human, they have accumulated more differences (Figure 1–47).

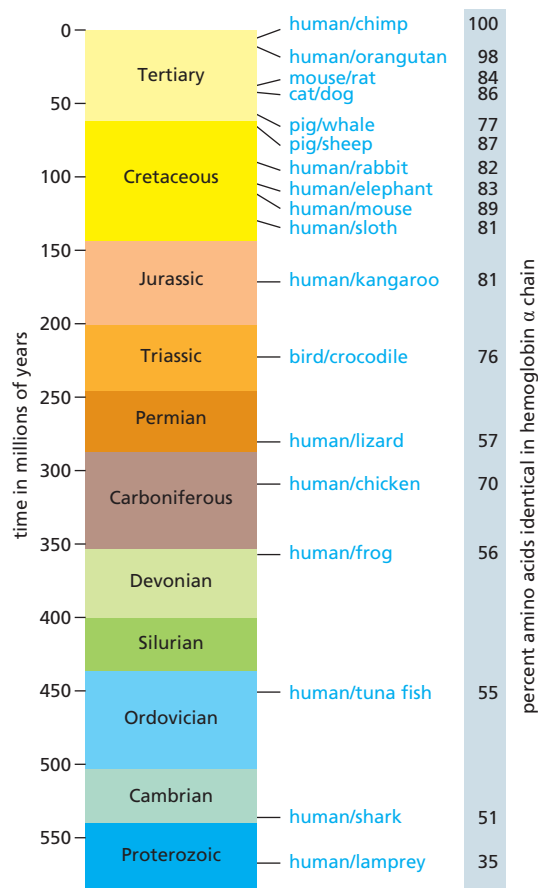


Figure 1–47 Times of divergence of different vertebrates. The scale on the left shows the estimated date and geological era of the last common ancestor of each specified pair of animals. Each time estimate is based on comparisons of the amino acid sequences of orthologous proteins; the longer the animals of a pair have had to evolve independently, the smaller the percentage of amino acids that remain identical. The time scale has been calibrated to match the fossil evidence showing that the last common ancestor of mammals and birds lived about 320 million years ago. The figures on the right show the amino acid sequence divergence for one particular protein—the α chain of hemoglobin. Note that although there is a clear general trend of increasing divergence with increasing time for this protein, there are irregularities that are thought to reflect the action of natural selection causing especially rapid changes in hemoglobin sequence when the organisms experienced special physiological demands. Some proteins that are subject to stricter functional constraints evolve much more slowly than hemoglobin, whereas others evolve as much as five times faster. (Adapted from S. Kumar and S.B. Hedges, *Nature* 392:917–920, 1998.)



Figure 1–48 Similar mutations produce the same effect in human and mouse. The human baby and the mouse shown here have remarkably similar abnormal white patches on their foreheads as a result of a mutation in the same gene (called *Kit*), which is required for the normal development, migration, and survival of some skin pigment cells. (Courtesy of R.A. Fleischman, from R.A. Fleischman et al., *Proc. Natl. Acad. Sci. USA* 88:10885–10889, 1991.)

The mouse, being small, hardy, and a rapid breeder, has become the foremost model organism for experimental studies of mammalian molecular cell biology. Many naturally occurring mutations are known, often mimicking the effects of corresponding mutations in humans to a remarkable extent (Figure 1–48). Moreover, methods have been developed to test the function of any chosen mouse gene or of any noncoding portion of the mouse genome by artificially creating mutations in the relevant part of the gene or genome, as we explain in Chapter 8.

Just one made-to-order mutant mouse can provide a wealth of information for the cell biologist. It reveals the effects of the chosen mutation in various contexts, simultaneously testing the action of the gene in the many different types of cells in the body that could in principle be affected. Studies of the mouse are so fundamental to understanding mammalian biology that we will encounter them in nearly every chapter of this book.

The COVID-19 Pandemic Has Focused Scientists on the SARS-CoV-2 Coronavirus

Having discussed several of the most prominent and well-studied model organisms—which are based on the cell as their fundamental unit—we now turn to an intensively studied virus. Viruses, which in essence feed on cells, are prevalent in all three domains of life: bacteria, archaea, and eukaryotes. We introduced them earlier in this chapter when we discussed several *E. coli* viruses that served as critical experimental systems for the initial development of molecular biology. Here, we focus on one prominent virus, SARS-CoV-2, that infects our own cells and has, due to the widespread attention it has received from scientists, become a model system for understanding eukaryotic viruses. But before discussing this virus in detail, we consider how viruses—genomes packaged in protective shells—first came to be, and how they have evolved over time.

As described in Chapter 6, cells are believed to have first evolved in an “RNA world,” before there were proteins or DNA molecules. Scientists suspect that even at that time, parasitic genetic elements were present, in the form of small RNA molecules that took advantage of more advanced replicating entities to proliferate. These are believed to have been the ancestors of today’s smallest viruses, which contain single-strand RNA genomes composed of as few as 3000 nucleotides. Thus, virus-like entities have probably been a ubiquitous feature of life on Earth for more than 3 billion years.

At a minimum, a virus requires a genome that encodes two core functions: first, a nucleic acid replication process that produces multiple copies of its genome once inside its host cell, and second, a genome-packaging process that surrounds these new genomes with a protective protein coat, while allowing the viruses to exit the host cell and subsequently enter others. But the viruses present today have evolved through billions of infectious cycles, during which there has been a constant war between host organisms and the viruses—with host cells evolving

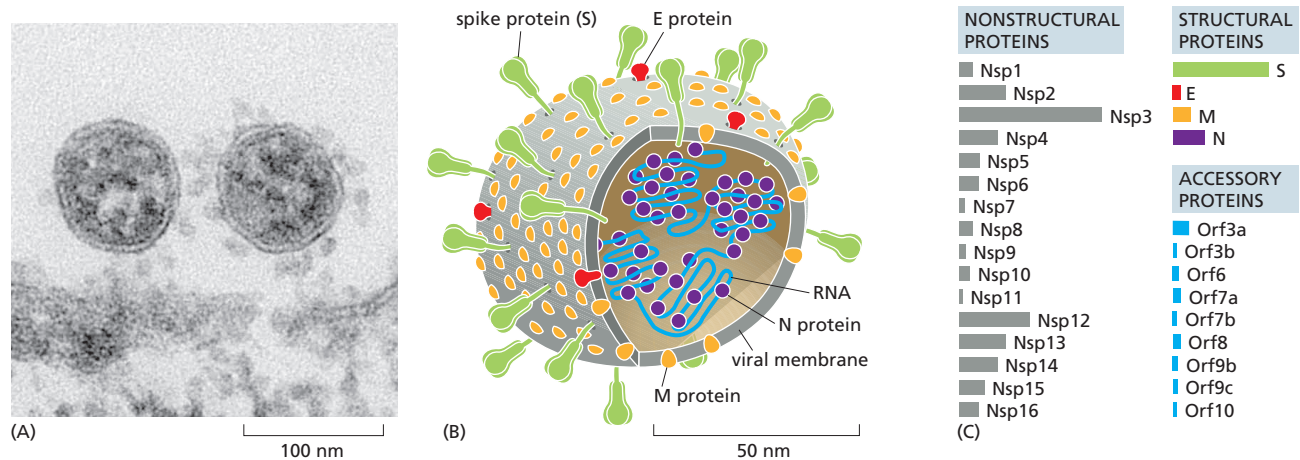


Figure 1-49 The coronavirus. (A) Electron micrograph of SARS-CoV-2 virus particles attached to the surface of a cultured monkey cell. (B) A cut-away drawing of the virus that highlights its protruding spike protein molecules plus a few other major proteins. The spike protein is the major target for vaccines that are designed to block infections, because it attaches the virus to the outside of host cells and then catalyzes transfer of the viral genome into the cell interior. As indicated, the RNA genome is packaged unevenly inside the enveloped virus particle. (C) The 29 proteins produced by SARS-CoV-2, grouped into three different categories. The locations of the structural proteins S, M, E, and N in the virus are indicated in panel B. Each of the proteins listed in the “accessory” category has a role in protecting the virus from host antiviral responses. The functions of the nonstructural proteins include binding to ribosomes to block host protein synthesis (Nsp1), forming a double-membrane “replication organelle” from host-cell membranes (Nsp 3, 4, and 6), and forming the RNA-dependent RNA polymerase (Nsp 7, 8, and 12). The way in which the virus reproduces itself, once inside a host cell, is shown in Figure 5-62. (A, from M. Laue et al. *Sci. Rep.* 11:3515, 2021. With permission from Cold Spring Harbor Press.)

multiple antivirus defenses and viruses evolving various ways to overcome these defenses. As a result, through cycles of random mutation followed by natural selection over long evolutionary times, most virus genomes have grown much larger than needed for their two core functions, with many of the additional genes encoding proteins that help the viruses to circumvent their host-cells’ defenses.

Coronavirus genomes are large, single-strand RNA molecules, about 30,000 nucleotides long. This RNA is packaged in a protein coat that is covered with a lipid bilayer envelope, from which protein spikes protrude (**Figure 1-49A and B**). Many coronavirus strains circulate in animal species, including pigs, birds, and bats. Some strains also circulate among humans; these so-called “endemic” strains cause only mild symptoms and are responsible for about one in four common colds. But on rare occasions, a bat coronavirus mutates in a way that allows it to infect humans, where it can cause very severe, even fatal, disease. It is thought that the COVID-19 pandemic of 2020 originated in this way.

The virus that causes COVID-19, SARS-CoV-2, produces 29 proteins (**Figure 1-49C**). Some are structural proteins that package the virus’s RNA genome into the virus particle. The nonstructural proteins are critical for replicating the viral genome inside of the host cell, as well as for ensuring that the viral genes are appropriately translated into proteins, including the viral RNA polymerase complex. And, as one would expect, other proteins help the virus to avoid the host’s immune defenses, which are described in Chapter 24.

The SARS-CoV-2 virus is closely related to the coronaviruses that cause colds, as well as to the SARS-CoV virus that emerged from bats in 2002 and killed nearly 1 in 10 of the humans it infected. We still do not understand what makes SARS-CoV and SARS-CoV-2 infections so much more dangerous to humans than the infections caused by their close relatives that cause only a mild cold. But, given the thousands of research laboratories currently focused on understanding the cell biology of SARS-CoV-2 with the aim of ameliorating the COVID-19 pandemic, we should know the answers to these questions in the near future. These studies are certain to make us much better prepared to deal with the next virus that emerges to threaten us.

Humans Are Unique in Reporting on Their Own Peculiarities

As humans, we have a special interest in the human genome. We want to know how our genes and their products work. But, even if you were a mouse, preoccupied with how mouse genes and their products work, humans would be attractive as model genetic organisms because of one special property: through medical examinations and self-reporting, we catalog our own genetic (and other) disorders. The human population is enormous, consisting today of some 8 billion individuals, and this self-documenting property means that a huge database exists of human mutations and their effects. And the human genome sequence of more than 3 billion nucleotide pairs has been determined for hundreds of thousands of people, making it easier than ever before to identify at a molecular level the precise genetic change responsible for any given human mutant phenotype.

But what precisely do we mean when we speak of the human genome? Whose genome? On average, any two people taken at random will differ at roughly 4 million different sites in their DNA sequence (see Table 4–3, p. 247). Thus, the human genome is very complex, embracing the entire pool of variant genes found in the human population. As described in Chapter 4, knowledge of this variation is helping us to understand human biology; for example, why some people are prone to one disease, others to another, and why some respond well to a drug, but others badly. It is also providing clues to our history, including population movements, interbreeding among our ancestors, the infections they suffered, and the diets they ate. All these things have left traces in the variant forms of genes that survive today in the human communities that populate our planet, and by exploiting this fact, scientists have been discovering fascinating aspects of our past.

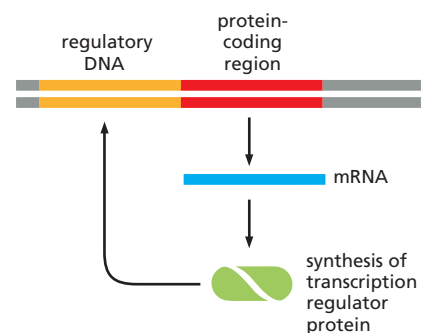
By drawing together the insights from humans, mice, fish, flies, worms, yeasts, plants, and bacteria—using DNA sequence similarities to map out the correspondences between one model organism and another—we are greatly enriching our understanding of them all.

To Understand Cells and Organisms Will Require Mathematics, Computers, and Quantitative Information

Empowered by knowledge of complete genome sequences, we can list the genes, proteins, and RNA molecules in a cell, and we have powerful methods to analyze the complex web of interactions between them. But how are we to use all this information to understand how cells work? Even for a single cell type belonging to a single species of organism, the current deluge of data seems overwhelming. The informal reasoning that biologists usually rely on seems increasingly inadequate in the face of such complexity.

The difficulty is more than just a matter of information overload. Biological systems are, for example, full of feedback loops, and the behavior of even the simplest of systems with feedback is remarkably difficult to predict by intuition alone (**Figure 1–50**); small changes in parameters can cause radical changes in outcome. To go from a circuit diagram to a prediction of the behavior of the

Figure 1–50 A very simple gene regulatory circuit. A single gene regulates its own expression because its protein product is a transcription regulator that binds to the regulatory DNA of its own gene. Simple schematic diagrams such as this are found throughout this book. They are often used to summarize what we know, but they leave many questions unanswered. When the protein binds, does it inhibit or stimulate transcription from the gene? How steeply does the transcription rate depend on the protein concentration? How long, on average, does a molecule of the protein remain bound to the DNA? How long does it take to make each molecule of mRNA or protein, and how quickly does each type of molecule get degraded? As explained in Chapter 8, mathematical modeling shows that we need quantitative answers to all these and other questions—obtained by direct observations and experiments—before we can predict the behavior of even this simple circuit. For different parameter values, the system may settle to a unique steady state; or it may behave as a switch, capable of existing in one or another of a set of alternative states; or it may oscillate; or it may even show large random fluctuations.



system, we need detailed quantitative information, and to draw deductions from that information we need mathematics and computers.

Such tools for quantitative reasoning are essential, but they are not all-powerful. You might think that, knowing how each protein in a cell influences each other protein, and how the expression of each gene is regulated by the products of other genes, we should soon be able to calculate how the cell as a whole will behave, just as an astronomer can calculate the orbits of the planets or a chemical engineer can calculate the flows through a chemical plant. But any attempt to perform this feat for anything close to an entire living cell rapidly reveals the limits of our present knowledge. The information we have, plentiful as it is, is full of gaps and uncertainties, and it is largely qualitative rather than quantitative. Most often, cell biologists studying a cell's control systems sum up their knowledge in simple schematic diagrams—this book is full of them—rather than in numbers, graphs, and differential equations.

To progress from qualitative descriptions and intuitive reasoning to quantitative descriptions and mathematical deduction is one of the biggest challenges for contemporary cell biology. So far, the challenge has been met for only a few very simple fragments of the machinery of living cells—subsystems involving a handful of different proteins, or two or three genes that regulate one another, where theory and experiment go closely hand in hand. We discuss some of these examples later in the book and devote much of Chapter 8 to some new approaches designed to answer the increasingly complex questions that arise in biology.

Knowledge and understanding bring the power to intervene—with humans, to prevent and treat disease; with plants, to create better crops; with bacteria, archaea, and fungi, to control them for our own benefit. All these biological enterprises are linked, because the genetic information of all living organisms is written in the same language. The recent ability of molecular biologists to read and decipher this language has already begun to transform our relationship to the living world. The account of cell biology in the subsequent chapters will, we hope, equip the reader to understand, and possibly to contribute to, the great biosciences adventure that we can anticipate through the rest of this century.

Summary

*Powerful new technologies, including rapid and cheap genome sequencing, are enabling rapid advances in our knowledge of human biology, with implications for understanding and treating human disease. But living systems are incredibly complex, and simpler model organisms have played a critical part in revealing universal genetic and molecular cell biological mechanisms. Thus, for example, early research on the bacterium *E. coli* and its viruses provided the foundations needed to decipher the fundamental genetic mechanisms in all cells. And research on the unicellular yeast *Saccharomyces cerevisiae*, which continues to serve as a simple model organism for eukaryotic cell biology, has revealed the molecular basis for many critical processes that have been strikingly conserved during more than a billion years of eukaryotic evolution. Biologists have also chosen a small number of multicellular organisms for intensive study: a worm, a fly, a fish, the mouse, and humans serve as model organisms for animals, and a small member of the cabbage family serves as a model for plant biology. Even today, research that focuses on these and other model organisms remains crucial for understanding ourselves, as well as for driving scientific and medical advances.*

PROBLEMS

Which statements are true? Explain why or why not.

1-1 DNA and RNA use the same four-letter alphabet.

1-2 Each member of the human hemoglobin gene family, which consists of seven genes arranged in two clusters on different chromosomes, is an ortholog to all of the other members.

1-3 Most of the DNA sequences in a bacterial genome code for proteins, whereas most of the DNA sequences in the human genome do not.

1-4 Without additional information, no amount of gazing at genome sequences will reveal the functions of genes.

Discuss the following problems.

1-5 “Life” is easy to recognize but difficult to define. Dictionaries commonly define life as “The state or quality that distinguishes living beings or organisms from dead ones and from inorganic matter, characterized chiefly by metabolism, growth, the ability to reproduce, and the ability to respond to stimuli.” Score a car, a cactus, and yourself with respect to these characteristics.

1-6 Since it was deciphered more than five decades ago, some have claimed that the genetic code must be a frozen accident, while others have argued that it was shaped by natural selection. A striking feature of the genetic code is its inherent resistance to the effects of mutation. For example, a change in the third position of a codon often specifies the same amino acid or one with similar chemical properties. The natural code resists mutation more effectively (is less susceptible to error) than most other possible versions, as illustrated in **Figure Q1-1**. Only one in a million computer-generated “random” codes is more error-resistant than the natural genetic code. Does the extraordinary mutation resistance of the genetic code argue in favor of its origin as a frozen accident or as a result of natural selection? Explain your reasoning.

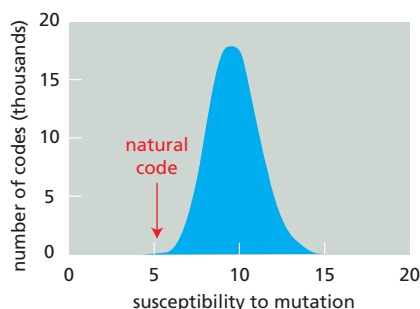


Figure Q1-1 Susceptibility to mutation of the natural code shown relative to that of millions of computer-generated alternative genetic codes (Problem 1-6). Susceptibility measures the average change in amino acid properties caused by random mutations in a genetic code. A small value indicates that mutations tend to cause minor changes. (Data courtesy of Steve Freeland.)

1-7 You have begun to characterize a sample obtained from the depths of the oceans on Europa, one of Jupiter’s moons. Much to your surprise, the sample contains a life-form that grows well in a rich broth. Your preliminary analysis shows that it is cellular and contains DNA, RNA, and protein. When you show your results to a colleague, she suggests that your sample was contaminated with an organism from Earth. What approaches might you try to distinguish between contamination and a novel cellular life-form that is based on DNA, RNA, and protein?

1-8 It is not so difficult to imagine what it means to feed on the organic molecules that living things produce. That is, after all, what we do. But what does it mean to “feed” on sunlight, as phototrophs do? Or, even stranger, to “feed” on rocks, as lithotrophs do? Where is the “food,” for example, in the mixture of chemicals (H_2S , H_2 , CO , Mn^{2+} , Fe^{2+} , Ni^{2+} , CH_4 , and NH_4^+) that spews from a hydrothermal vent?

1-9 How many possible different trees (branching patterns) can be drawn to display the evolution of bacteria, archaea, and eukaryotes, assuming that they all arose from a common ancestor?

1-10 The genes for ribosomal RNA are highly conserved (relatively few sequence changes) in all organisms on Earth; thus, they have evolved very slowly over time. Were ribosomal RNA genes “born” perfect?

1-11 Rates of evolution appear to vary in different lineages. For example, the rate of evolution in the rat lineage is significantly higher than in the human lineage. These rate differences are apparent whether one looks at changes in nucleotide sequences that encode proteins and are subject to selective pressure or at changes in noncoding nucleotide sequences, which are not under obvious selection pressure. Can you offer one or more possible explanations for the slower rate of evolutionary change in the human lineage versus the rat lineage?

1-12 Genes participating in informational processes such as replication, transcription, and translation undergo horizontal gene transfer between species much less often than do genes involved in metabolism. The basis for this inequality is unclear at present, but one suggestion is that it relates to the underlying complexity of the two types of processes. Informational processes tend to involve large aggregates of different gene products, whereas metabolic reactions are usually catalyzed by enzymes composed of a single protein. Why would the complexity of the underlying process—informational or metabolic—have any effect on the rate of horizontal gene transfer?

1-13 Animal cells have neither cell walls nor chloroplasts, whereas plant cells have both. Fungal cells are somewhere in between; they have cell walls but lack chloroplasts. Are fungal cells more likely to be animal cells that

gained the ability to make cell walls or to be plant cells that lost their chloroplasts? This question represented a difficult issue for early investigators who sought to assign evolutionary relationships solely on the basis of cell characteristics and morphology. How do you suppose that this question was eventually decided?

1–14 *Giardia lamblia* is a fascinating eukaryotic parasite; it contains a nucleus but no mitochondria and no discernible endoplasmic reticulum or Golgi apparatus—one of the very rare examples of such a cellular organization among eukaryotes. This cell organization might have arisen because *Giardia* is an ancient lineage that separated from the rest of the eukaryotes before mitochondria were acquired and internal membranes were developed. Or it might be a stripped-down version of a more standard eukaryote that has lost these structures because they are not necessary for its parasitic lifestyle. How might you use nucleotide sequence comparisons to distinguish between these alternatives?

1–15 When plant hemoglobin genes were first discovered in legumes, it was so surprising to find a gene typical of animal blood that it was hypothesized that the plant gene arose by horizontal transfer from an animal. Many more hemoglobin genes have now been sequenced, and a phylogenetic tree based on some of these sequences is shown in **Figure Q1–2**.

A. Does this tree support or refute the hypothesis that the plant hemoglobins arose by horizontal gene transfer?

B. Supposing that the plant hemoglobin genes were originally derived from a parasitic nematode, for example, what would you expect the phylogenetic tree to look like?

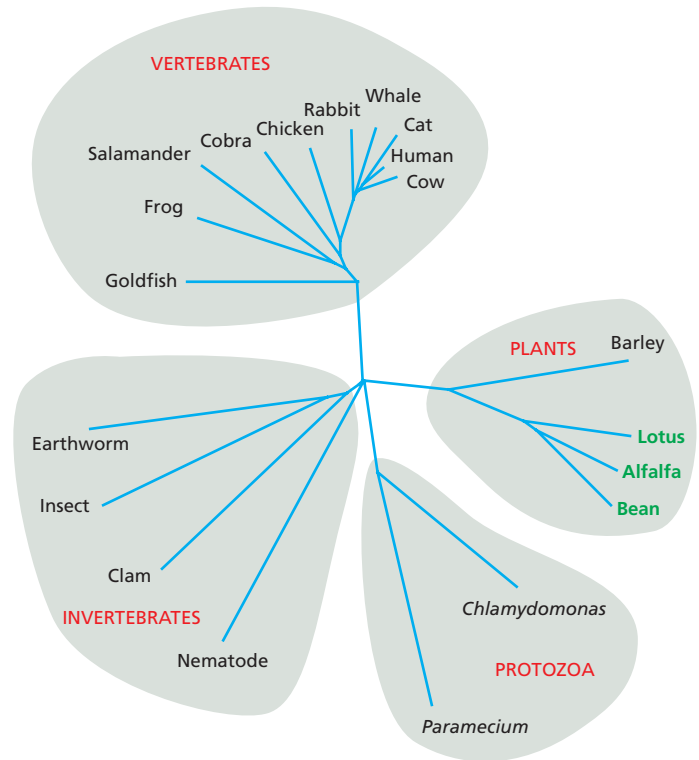


Figure Q1–2 Phylogenetic tree for hemoglobin genes from a variety of species (Problem 1–15). The legumes are highlighted in green. The lengths of lines that connect the present-day species represent the evolutionary distances that separate them.

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